

An ultrasensitive method for detection of cell-free RNA

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Sensitive methods for detection of cell-free RNA (cfRNA) could facilitate non-invasive gene expression profiling and monitoring of diseases^{1–6}. Here we describe RARE-seq (random priming and affinity capture of cfRNA fragments for enrichment analysis by sequencing), a method optimized for cfRNA analysis. We demonstrate that platelet contamination can substantially confound cfRNA analyses and develop an approach to overcome it. In analytical validations, we find RARE-seq to be approximately 50-fold more sensitive for detecting tumour-derived cfRNA than whole-transcriptome RNA sequencing (RNA-seq), with a limit of detection of 0.05%. To explore clinical utility, we profiled 437 plasma samples from 369 individuals with cancer or non-malignant conditions and controls. Detection of non-small-cell lung cancer expression signatures in cfRNA increased with stage (6 out of 20 (30%) in stage I; 5 out of 8 (63%) in stage II; 10 out of 15 (67%) in stage III; 80 out of 96 (83% sensitivity) in stage IV at 95% specificity) and RARE-seq was more sensitive than tumour-naïve circulating tumour DNA (ctDNA) analysis. In patients with *EGFR*-mutant non-small-cell lung cancer who developed resistance to tyrosine kinase inhibitors, we detected both histological transformation and mutation-based resistance mechanisms. Finally, we demonstrate the potential utility of RARE-seq for determination of tissue of origin, assessing benign pulmonary conditions and tracking response to mRNA vaccines. These results highlight the potential value of ultrasensitive cfRNA analysis and provide proof of concept for diverse clinical applications.

Blood-based liquid biopsies enable non-invasive characterization of diverse cancers with existing applications including early detection, tissue-of-origin (TOO) classification, genotyping and disease monitoring. These applications have largely focused on ctDNA, for example, as routinely used clinically for non-invasive tumour genotyping of actionable mutations in patients with advanced malignancies. cfRNA analysis could potentially complement ctDNA analysis and allow

broader molecular characterization of cancers^{1–6}. However, the vast majority of circulating transcripts are shed by haematopoietic cells^{7–10}, making detection of rare circulating tumour RNA (ctRNA) molecules challenging.

Here we describe RARE-seq, a method specifically optimized for cfRNA analysis. We demonstrate that this approach improves the limit of detection for ctRNA over previous methods while maintaining high

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specificity for tumour-naïve cancer detection. We further explore the potential utility of RARE-seq in diverse clinical applications, including non-invasive ctRNA genotyping, treatment resistance monitoring, TOO analysis and molecular characterization of non-malignant conditions.

Pre-analytical factors affecting cfRNA

As previously described^{1,8,11}, we observed that cfRNA is highly fragmented and does not contain detectable 18S and 28S rRNA peaks (Extended Data Fig. 1a), indicating high levels of degradation. In blood samples from healthy donors with no history of cancer, the median concentration of cfRNA was 220 pg per millilitre plasma, representing the equivalent of approximately 20 cells ($n = 117$; Extended Data Fig. 1b). We therefore began by evaluating factors that affected cfRNA recovery and downstream expression analysis (Supplementary Note). We identified multiple blood collection and RNA extraction protocols that allowed similar RNA recovery from plasma (Extended Data Fig. 1c,d) and found that factors such as haemolysis were not associated with cfRNA yields (Extended Data Fig. 1e). When investigating the effects of sample storage, we observed a decrease of cfRNA concentration during the first month of freezer storage ($P = 0.003$; Extended Data Fig. 1f), but no significant effect after the first month for up to 10 years ($P = 0.74$; Extended Data Fig. 1g). We next optimized multiple steps in the assay workflow, including removal of contaminating DNA, complementary DNA (cDNA) synthesis and end repair. These optimizations enhanced library preparation efficiency from low cfRNA inputs and minimized effects of contaminating cfDNA (Extended Data Fig. 2a–h). We further confirmed that our optimizations resulted in reproducible cfRNA expression profiles in healthy controls (Extended Data Fig. 2i–k). Collectively, we refer to the optimized method as RARE-seq (Fig. 1a).

Platelet genes are enriched in cfRNA

We next characterized expression differences between cellular and cfRNA by applying RARE-seq to cfRNA and matched leukocyte RNA from ten healthy adults. Differential expression analysis identified 8,336 significant genes, including 5,271 genes over-represented in cfRNA and 3,065 genes over-represented in leukocyte RNA (Fig. 1b). Using gene set enrichment analysis (GSEA) and cell-type-specific gene sets from a single-cell RNA-seq atlas¹² (Fig. 1c), we found that cfRNA was significantly enriched for non-haematopoietic cell types, including endothelial cells, hepatocytes and fibroblasts ($n = 71$ total), consistent with previous work¹⁰. By contrast, leukocyte RNA was significantly enriched for 12 major haematopoietic cell types of myeloid and lymphoid lineages, including T cells, natural killer cells, neutrophils and monocytes. Only two haematopoietic cell types were enriched in cfRNA – erythroid precursors and platelets – suggesting their contributions to the cfRNA pool during erythropoiesis and megakaryocytopoiesis.

Given the significant enrichment of platelet transcripts in cfRNA (Fig. 1c), we suspected that residual platelets not removed during plasma isolation may confound cfRNA analysis. To test this, we compared cfRNA concentrations using three different centrifugation speeds to isolate plasma ($n = 32$). Consistent with our hypothesis, the median concentration was approximately 7.5-fold higher at 1,200g than at 2,500g (1.72 ng ml^{-1} versus 0.23 ng ml^{-1} ; $P = 0.000052$; Fig. 1d). Furthermore, average expression of platelet-specific genes ($n = 130$)¹² was significantly associated with centrifugation speed (Fig. 1e). Principal component analysis revealed that the first principal component (PC1) captured 70% of expression variation and was strongly associated with platelet expression ($R = -0.99$, $P \leq 2.2 \times 10^{-16}$; Fig. 1f,g). Similar results were found even among samples processed with a fixed centrifugation speed, suggesting that differences in platelet counts may also contribute to variation of platelet transcripts in cfRNA ($R = 0.98$, $P \leq 1.3 \times 10^{-11}$; Fig. 1h). Collectively, these results suggest that platelets represent the largest source of expression variation in cfRNA and that

platelet contamination is a critical pre-analytical variable to control for in cfRNA analyses.

Although individual platelets contain less RNA content than leukocytes¹³, platelets outnumber leukocytes by approximately two orders of magnitude in blood. In addition, physiological variation in platelet counts span an approximately threefold range in healthy adults¹⁴ with further variability observed in the context of cancer¹⁵. Therefore, even optimized methods to minimize plasma contamination by platelets could fail to completely remove platelet-derived RNA from cfRNA samples. Separately, from a practical standpoint, it is often not feasible to standardize blood banking protocols, especially when profiling previously collected samples. Therefore, we investigated whether expression contributions from platelet contamination could be algorithmically eliminated. To do so, we developed an *in silico* approach to remove unwanted gene expression driven by varying platelet contamination levels (Fig. 1i). This approach aims to separate wanted variation (that is, cancer-associated gene expression differences) from unwanted variation (that is, technical or biological noise) by performing factor analysis of expression from negative control genes for which only unwanted variation is expected to be present (Methods). We found that this approach successfully removed the unwanted variation contributed by platelets, as shown by the lack of the correlation between platelet expression and PC1 after correction ($R = 0.07$, $P = 0.69$; Fig. 1j). Hierarchical clustering further confirmed this effect, as samples clustered according to platelet gene expression before correction (Fig. 1k), but according to inter-individual expression differences after correction (Fig. 1l). Therefore, addressing platelet contamination in cfRNA analyses is critical for reliably uncovering gene expression contributions from non-platelet sources.

Enhancing sensitivity of ctRNA detection

Although cfRNA is highly enriched for expression of genes from non-haematopoietic sources, the absolute expression levels of these genes are relatively low compared with haematopoietic-derived transcripts (for example, globins) and highly expressed housekeeping genes (for example, β -actin). We therefore hypothesized that selectively capturing genes absent from or lowly expressed in healthy control cfRNA could enhance detection of non-haematopoietic tissue or disease signatures and expand the utility of RARE-seq to a wide range of clinical applications. To test this, we identified such ‘rare abundance genes’ (RAGs) using whole-coding transcriptome RARE-seq data from healthy cfRNA samples ($n = 50$) and RNA-seq data from healthy whole-blood samples ($n = 307$) from the GTEx project¹⁶ (Fig. 2a). We defined RAGs based on consistent low or absent expression in cfRNA and whole blood ($n = 5,632$ genes; Extended Data Fig. 3a and Supplementary Table 1). As compared with all protein-coding transcripts, RAGs were substantially enriched for both non-haematopoietic tissue-specific genes and cancer-specific genes from diverse cancers (Fig. 2b,c).

The enrichment of genes specific for multiple tissue and cancer types among RAGs suggests that a RAG-targeted sequencing panel could increase sensitivity for detection of cancer or other conditions of interest without requiring a bespoke panel design for each application. We therefore designed a capture panel targeting 4,737 RAGs and 50 housekeeping genes, prioritizing genes with relatively uniform expression across healthy cfRNA and whole-blood samples while covering a broad dynamic range of expression (Extended Data Fig. 3b). Although these modules comprise the core of the RAG-focused capture panel design and could be used by themselves to explore diverse pathophysiological conditions without further customization, additional targets may be optionally added to the capture panel if desired (Extended Data Fig. 3c). For example, for the purposes of this study, we added genes known to be recurrently mutated in lung cancers ($n = 122$ genes), including genes such as *TP53*, *KRAS*, and *RET* (Extended Data Fig. 3d; Methods). The full collection of captured genes is listed in Supplementary Table 2.

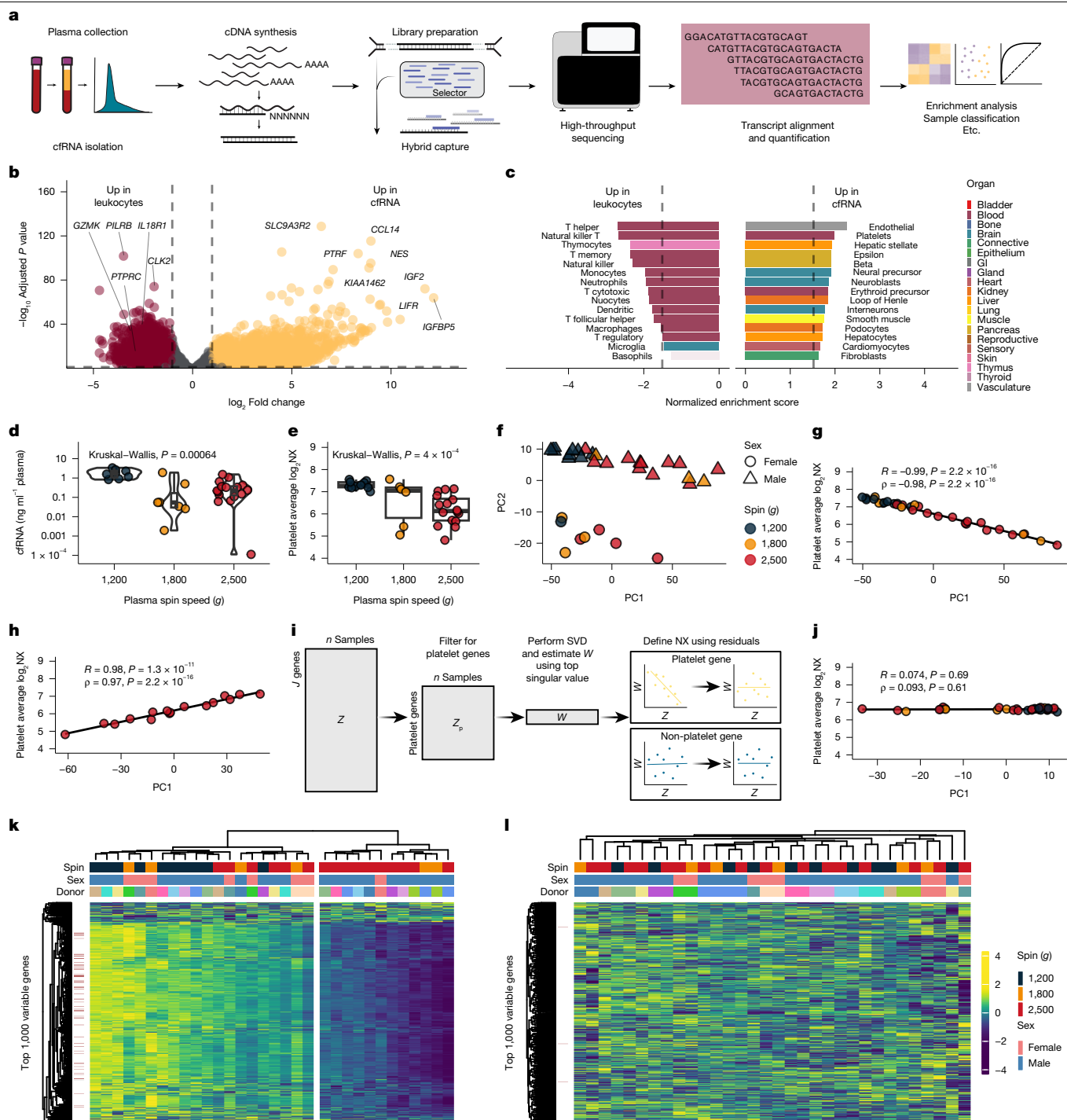


Fig. 1 | Platelet and non-haematopoietic transcripts are enriched in cfRNA.

a, Schematic of the RARE-seq method. The schematic was adapted from ref. 36, Springer Nature. **b**, Differential expression analysis comparing cfRNA and matched leukocyte RNA from healthy donors ($n = 10$ pairs). Significance was determined using Wald test and adjusted using Benjamini–Hochberg correction. **c**, GSEA using cell-type-specific signatures from PanglaoDB¹² ($n = 170$ cell types). The top 15 positively and negatively enriched cell types are shown, and the dashed line delineates significant enrichment. GI, gastrointestinal. **d**, Relationship between centrifugation speed during plasma isolation and cfRNA concentration per millilitre plasma ($n = 10$ at 1,200g, $n = 6$ at 1,800g and $n = 16$ at 2,500g). Comparisons were performed using the Kruskal–Wallis test. **e**, Relationship between centrifugation speed and platelet-specific gene expression in cfRNA samples from panel **d**. Comparisons were performed using the Kruskal–Wallis test. Average $\log_2$ NX, mean of normalized expression. **f**, Principal component analysis of cfRNA expression clusters according to spin

speed (PC1) and sex (PC2). The top 1,000 variable genes in cfRNA samples from panel **d** were used for principal component analysis. **g**, Association between average platelet-specific gene expression and PC1 in cfRNA samples from panel **d** before correction. Pearson and Spearman correlation are shown. **h**, Association between average platelet-specific gene expression and PC1 when considering only samples processed using the same centrifugation speed ($n = 16$ at 2,500g) before correction. **i**, Schematic of the platelet-directed correction approach (Methods). SVD, singular value decomposition; W , platelet correction factor; Z , z scores; Z_p , z scores of platelet genes. **j**, Association between platelet-specific gene expression and PC1 after platelet-directed correction. **k**, Hierarchical clustering of the top 1,000 variable genes in cfRNA before correction. Clustering was performed using Euclidean distance. **l**, Hierarchical clustering of the top 1,000 variable genes after platelet-directed correction. In the boxplots, the centre line denotes the median, the box contains the interquartile range (IQR), and the whiskers denote extrema no farther than $1.5 \times$ IQR.

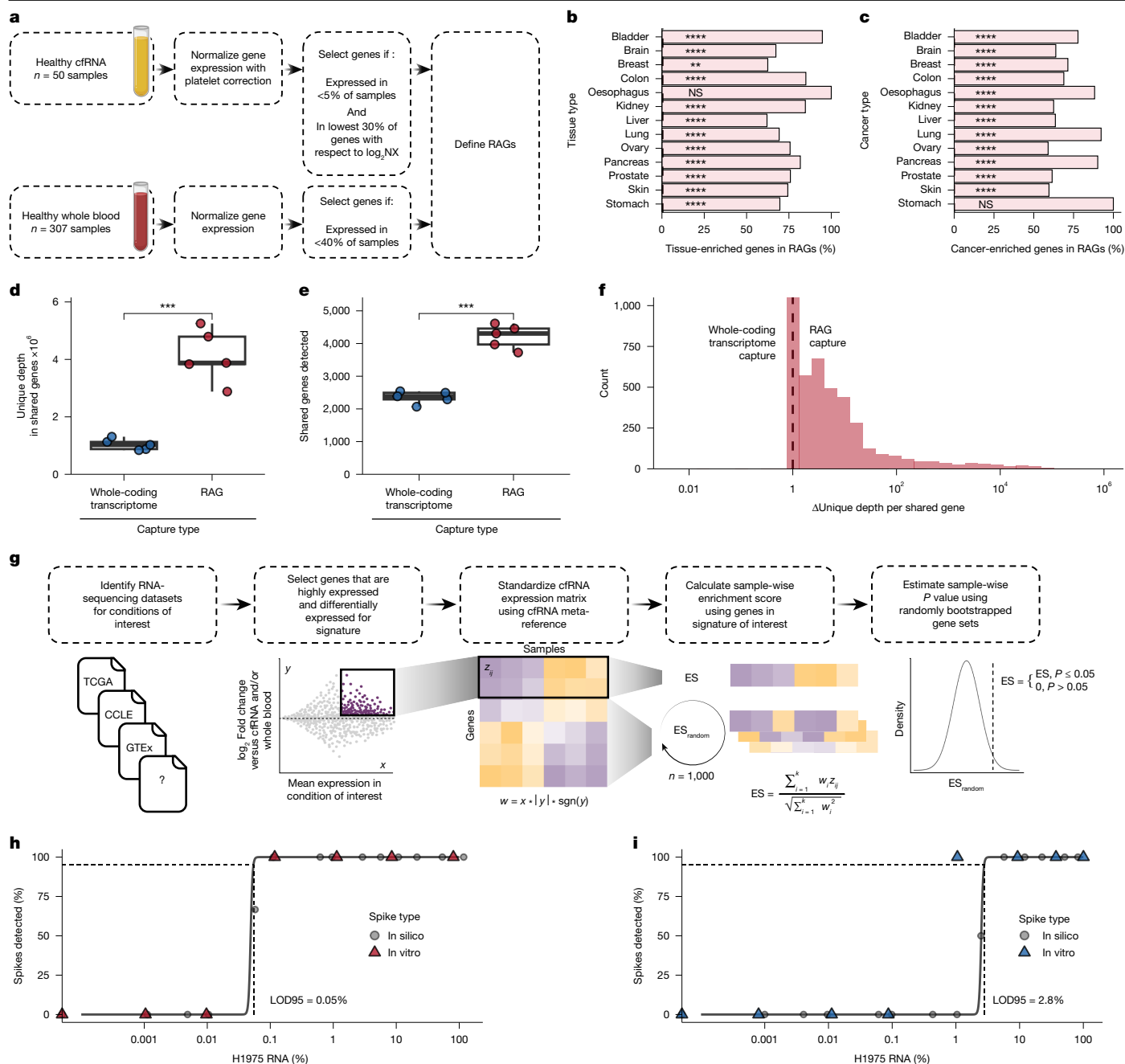


Fig. 2 | Targeting transcripts absent from healthy cfRNA improves analytical sensitivity for lung cancer detection. **a**, Schematic depicting the selection of RAGs in plasma. The illustrations of the plasma and blood tubes were created using BioRender (<https://biorender.com>). **b,c**, Fraction of tissue-enriched (**b**) and cancer-enriched (**c**) genes that overlap RAGs. Tissue-enriched and cancer-enriched genes were selected as defined in the Methods. Comparisons were performed using two-sided Fisher's exact test and adjusted with Benjamini-Hochberg correction (** $P < 0.01$, **** $P < 0.0001$). NS, not significant. **d**, Unique depth in libraries captured using either whole-coding transcriptome or RAG capture panels ($n = 5$ matched libraries). Libraries were downsampled to have equivalent sequencing depth and filtered to include regions captured by both panels. Comparisons were performed using paired two-sided Student's t -test (*** $P = 0.0009$). **e**, Number of shared genes detected in libraries in panel **d**. The threshold for detection was one or more unique reads. Comparisons were performed using paired two-sided Student's t -test (*** $P = 0.0008$). **f**, Histogram

depicting the difference in unique depth per gene for libraries in panel **d**. The dashed line depicts equivalent depth between capture panels. **g**, Schematic of cfRNA enrichment score (ES) analytical framework (see Methods). The data matrices, scatter plot and arrows were created using BioRender (<https://biorender.com>). **h**, Relationship between cfRNA enrichment score detection rate and lung cancer cell line (NCI-H1975) fraction for spikes captured with the RAG panel ($n = 3$ replicates per spike). The presence of cancer RNA was detected in each spike using an NCI-H1975 cfRNA enrichment score. Logistic regression was used to estimate the LOD95 (shown as dashed line) from in silico spikes (shown as grey circles). Results from in vitro spikes are shown as red triangles. **i**, Empirical LOD95 of NCI-H1975 spikes captured using the whole-coding transcriptome panel ($n = 1$ replicate per spike) evaluated using the same approach as panel **h**. In the boxplots, the centre line denotes the median, the box contains the IQR, and the whiskers denote extrema no farther than $1.5 \times$ IQR.

We next profiled five healthy cfRNA samples using both whole-coding transcriptome and RAG capture panels using RARE-seq. Gene expression was highly correlated between matched libraries ($R = 0.98$,

$P \leq 2.2 \times 10^{-16}$; Extended Data Fig. 3e), including for the housekeeping genes ($R = 0.97$, $P \leq 2.2 \times 10^{-16}$; Extended Data Fig. 3f). Within the subset of genes shared across both capture panels, RAG capture resulted in

higher unique sequencing depth (Fig. 2d), more genes detected (Fig. 2e) and higher unique sequencing depth per gene detected (Fig. 2f). Thus, targeted capture of RAGs effectively increases sensitivity for detection of genes that are lowly expressed in healthy cfRNA.

We next set out to determine the limit of 95% detection (LOD95) for transcriptional signatures of interest, using either whole-coding transcriptome or RAG capture panels. To score the presence of a gene signature of interest in cfRNA samples, we developed an analytical framework to compute a cfRNA gene signature enrichment score by comparing the expression of pre-defined signature genes in a sample of interest to a 'meta-reference' consisting of cfRNA samples from healthy controls and using bootstrapping to estimate its significance (Fig. 2g; Methods). An important advantage of this approach is that unlike machine learning-based methods, it does not require a large training cohort of patient cfRNA samples, as it leverages existing gene expression datasets to define gene signatures and associated gene-wise weights based on their expression levels in the tissue or condition of interest.

To establish the LOD95 of RARE-seq, we generated in silico serial dilutions of RNA from the NCI-H1975 NSCLC cell line into healthy control cfRNA at various cancer fractions. To enable using our cfRNA enrichment score framework, we generated an NCI-H1975-specific gene signature, comprising the top differentially expressed genes (DEGs) distinguishing NCI-H1975 from a meta-reference healthy cfRNA profile (H1975 Sig; $n = 122$ genes). Using this signature, the LOD95 for RAG RARE-seq was 0.05% (Fig. 2h), more than 50-fold more sensitive than for whole-coding transcriptome RARE-seq (2.8%; Fig. 2i), demonstrating the value of targeting RAGs. To experimentally confirm this result, we created in vitro dilution series, by spiking NCI-H1975 RNA into healthy donor cfRNA at defined concentrations. These in vitro spikes demonstrated strong agreement with our in silico results (Fig. 2h,i). We also evaluated the effect of sequencing coverage depth (range of 10–50 million reads) on the LOD95 of RAG-targeted RARE-seq and found that higher sequencing depth improved the LOD95 (Extended Data Fig. 3i). Accordingly, we targeted 50 million reads for subsequent experiments using RARE-seq with the RAG panel.

Detection of non-small-cell lung cancer

Having developed and optimized RARE-seq, we focused the remainder of our study on generating proof-of-concept data for cfRNA analysis across several clinical applications. Given the higher analytic sensitivity observed by capturing RAGs, we used RAG-focused RARE-seq for these experiments. First, we investigated performance for tumour-naïve detection of lung adenocarcinomas (LUADs), the most common type of non-small-cell lung cancer (NSCLC). We analysed cfRNA from 139 patients with LUAD ($n = 20$ patients with stage I, $n = 8$ patients with stage II, $n = 15$ patients with stage III and $n = 96$ patients with stage IV) and 50 controls without cancer, including 26 risk-matched controls with a history of substantial tobacco exposure collected at the time of low-dose computed tomography (LDCT) screening for lung cancer. Concentrations of cfRNA and sequencing depth were similar between groups (Extended Data Fig. 4a,b). Platelet-adjusted differential expression analysis between LUADs and controls identified 96 genes overexpressed in LUADs, including key genes known to be highly expressed in LUADs such as *SLC34A2*, *SFTA2* and *NKX2-1* (also known as *TTFI*; Fig. 3a). LUAD DEGs were confirmed to be highly expressed in LUAD tumours from The Cancer Genome Atlas (TCGA)¹⁷, whereas were depleted in meta-reference cfRNA controls (Fig. 3b). In addition, using cell-type-specific gene signatures, GSEA revealed LUAD cfRNA to be most enriched for epithelial cells including pulmonary alveolar type II cells, and most depleted for naïve and memory B lymphocytes (Fig. 3c). Thus, cfRNA from patients with LUAD is enriched for transcripts that are highly expressed in LUAD tumours.

To further quantify the enrichment of LUAD-derived transcripts, we applied the cfRNA enrichment score framework developed above

using a LUAD-specific gene signature (LUAD Sig; $n = 72$ genes) identified using RNA-seq data from LUAD tumours within TCGA¹⁷, LUAD cell lines from the Cancer Cell Line Encyclopedia (CCLE)¹⁸, whole blood (GTEx)¹⁶ and meta-reference cfRNA controls (Methods). The LUAD Sig was detected in 101 out of 139 participants with LUAD (73% sensitivity at 95% specificity in non-cancer controls; Fig. 3d and Supplementary Table 3) and was significantly associated with stage (for sensitivity by stage, I = 30%, II = 62%, III = 67% and IV = 83%; $P = 9.1 \times 10^{-16}$; Fig. 3e). In addition, in a subset of plasma samples that were also interrogated using tumour-naïve ctDNA analysis ($n = 38$; Methods), the LUAD cfRNA enrichment score was significantly correlated with ctDNA variant allele frequency ($R = 0.43$, $P = 0.009$; Fig. 3f). Of these samples, 34% was detected by only RARE-seq and none was detected by only ctDNA analysis (Fig. 3g), indicating that tumour-naïve ctRNA analysis is more sensitive than tumour-naïve ctDNA analysis. Technical variables that varied across groups, such as library input and unique sequencing depth, extraction method and capture panel, were not associated with sensitivity (Extended Data Fig. 4e–h), indicating that these variables did not drive detection. In addition, detection was not significantly associated with biological factors such as patient smoking status or the presence of oncogenic driver mutations (Extended Data Fig. 4i–k). Finally, we found that platelet correction significantly improved sensitivity ($P = 0.03$; Extended Data Fig. 4l).

Genotyping of somatic mutations in ctRNA

We also explored the ability to genotype cancer-derived somatic mutations using RARE-seq. Although non-invasive tumour genotyping is already routinely performed clinically using ctDNA and ctRNA-based genotyping is unlikely to outperform it, variant detection could be a useful addition to cfRNA analysis. We therefore developed a custom ctRNA genotyping approach to detect clinically actionable variants, including single-nucleotide variants (SNVs), insertions or deletions (indels), splice variants and gene fusions, in recurrent NSCLC oncogenes. To ensure specificity, putative variants were censored if their variant allele frequency did not significantly exceed the background error rate observed in a subset of withheld controls ($n = 50$; Extended Data Fig. 5a; Methods). Performance was assessed in 136 patients with stage IV LUAD with known variants from previous tumour or ctDNA genotyping, as well as 97 healthy and LDCT controls (Supplementary Table 4). We detected one or more somatic driver mutations in 28% of patients with LUAD and 1% of controls, including key NSCLC oncogenic drivers such as L858R and exon 19 deletions in *EGFR*, G12C in *KRAS*, *ROS1* and *RET* fusions, and *MET* exon skipping events (Fig. 3h,i and Extended Data Fig. 5b–d). Of the 162 variants identified by tumour tissue genomic DNA or ctDNA genotyping in the same patients, 35 (22%) were also detectable in ctRNA, and detection rates were similar across variant types (Fig. 3j). Of the 34 variants identified by tumour tissue genomic DNA in patients with matched cfRNA and cfDNA data available, 11 variants were detectable in ctDNA compared with 3 variants detectable in ctRNA, indicating that ctRNA genotyping may be useful to supplement expression-based analyses but should not replace ctDNA genotyping (Extended Data Fig. 5e). We observed a significant relationship between variant detection rates and corresponding expression levels; for instance, *EGFR* variants were more frequently detected in cfRNA samples with high *EGFR* expression ($P = 0.0001$; Fig. 3k). Thus, RARE-seq analysis allows simultaneous identification of somatic mutations in patients with cancer.

Given that ctRNA expression signatures and ctRNA somatic variation could individually distinguish between patients with cancer and controls without cancer, we next investigated whether a machine learning approach could combine these features into a single detection algorithm. We trained a weighted elastic net classifier to distinguish LUAD and control cfRNA using tenfold nested cross-validation in a training cohort of 96 stage IV cases and 47 controls. The model demonstrated

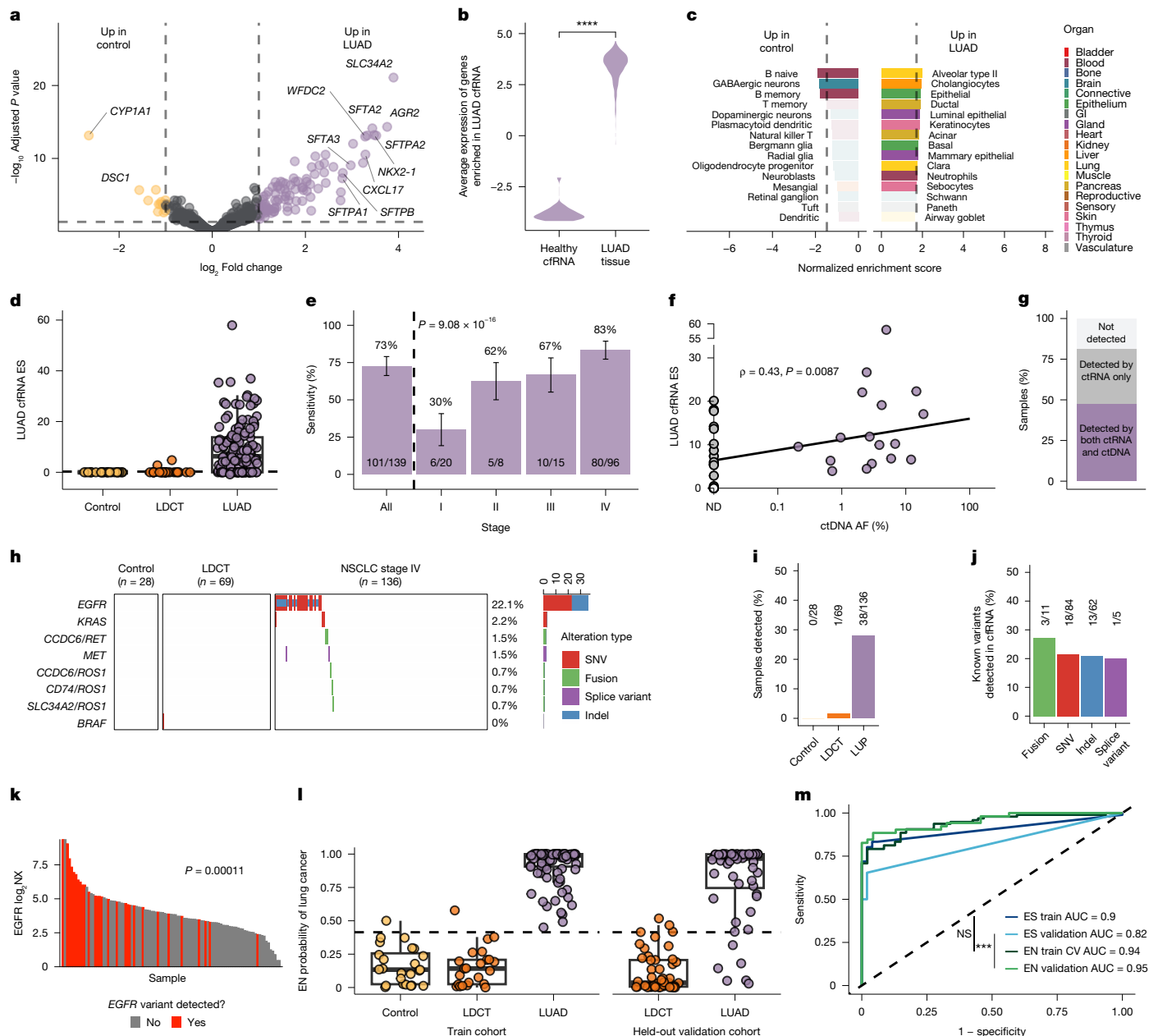


Fig. 3 | Detection of ctRNA in plasma from patients with NSCLC.

a, Differential expression between LUAD cfRNA ($n = 139$) and non-cancer cfRNA ($n = 50$). Significance was determined using Wald test and Benjamini–Hochberg correction. **b**, Average expression of genes enriched in LUAD cfRNA in LUAD tumours ($n = 385$) and meta-reference cfRNA ($n = 15$). Comparison used two-sided Wilcoxon test ($****P \leq 2.2 \times 10^{-16}$). **c**, GSEA using PanglaoDB¹² cell-type signatures. Top 15 positive and negative types are shown, and the dashed line delineates significant enrichment. **d**, Enrichment scores for the LUAD Sig ($n = 72$ genes) in cfRNA from patients with LUAD ($n = 139$), risk-matched controls ($n = 26$) and healthy controls ($n = 24$). **e**, Sensitivity of LUAD cfRNA enrichment score detection, summarized by stage. Detection threshold achieving 95% or more specificity was used. Error bars represent the 95% binomial confidence interval. Stages were compared using Spearman correlation. **f**, Relationship between LUAD cfRNA enrichment score and ctDNA allele frequency (AF) in matched plasma ($n = 38$). ND, not detected. **g**, Percentage of matched samples

with detectable ctRNA and/or ctDNA. **h**, OncoPrint of variants detected in cfRNA from patients with stage IV LUAD with one or more tissue-adjudicated or ctDNA-adjudicated variants and controls. **i**, Percentage of samples with one or more variants detected in cfRNA. **j**, Percentage of known variants detected in cfRNA. **k**, Relationship between *EGFR* expression and variant detection for samples with known *EGFR* variants. **l**, Probability of lung cancer from the elastic net (EN) model (for training, $n = 96$ LUAD and 47 control; for validation, $n = 52$ LUAD and 46 control). Detection threshold achieving 95% or more specificity in the training cohort was used (shown as dashed line). **m**, Receiver operating characteristic curves summarizing detection using LUAD cfRNA enrichment score and elastic net probabilities. Area under the curve (AUC) was compared with DeLong's test. CV, cross-validation. In the boxplots, the centre line denotes the median, the box contains the IQR, and the whiskers denote extrema no farther than 1.5 $\times$ IQR.

high classification performance (area under receiver operating characteristic curve, AUROC = 0.94; Fig. 3l,m and Supplementary Table 5) and selected 23 features including gene expression features (for example, key LUAD markers such as *SLC34A2*) and mutation-based features (that is, presence of L858R in *EGFR* or a *KRAS* mutation; Extended Data

Fig. 5f). To test whether our cohort was sufficiently large, we repeated classifier training and cross-validation using sub-sampling and found stable performance when using at least 100 samples for model training (Extended Data Fig. 5g). The elastic net model performed similarly well in a withheld validation cohort collected at an independent institution

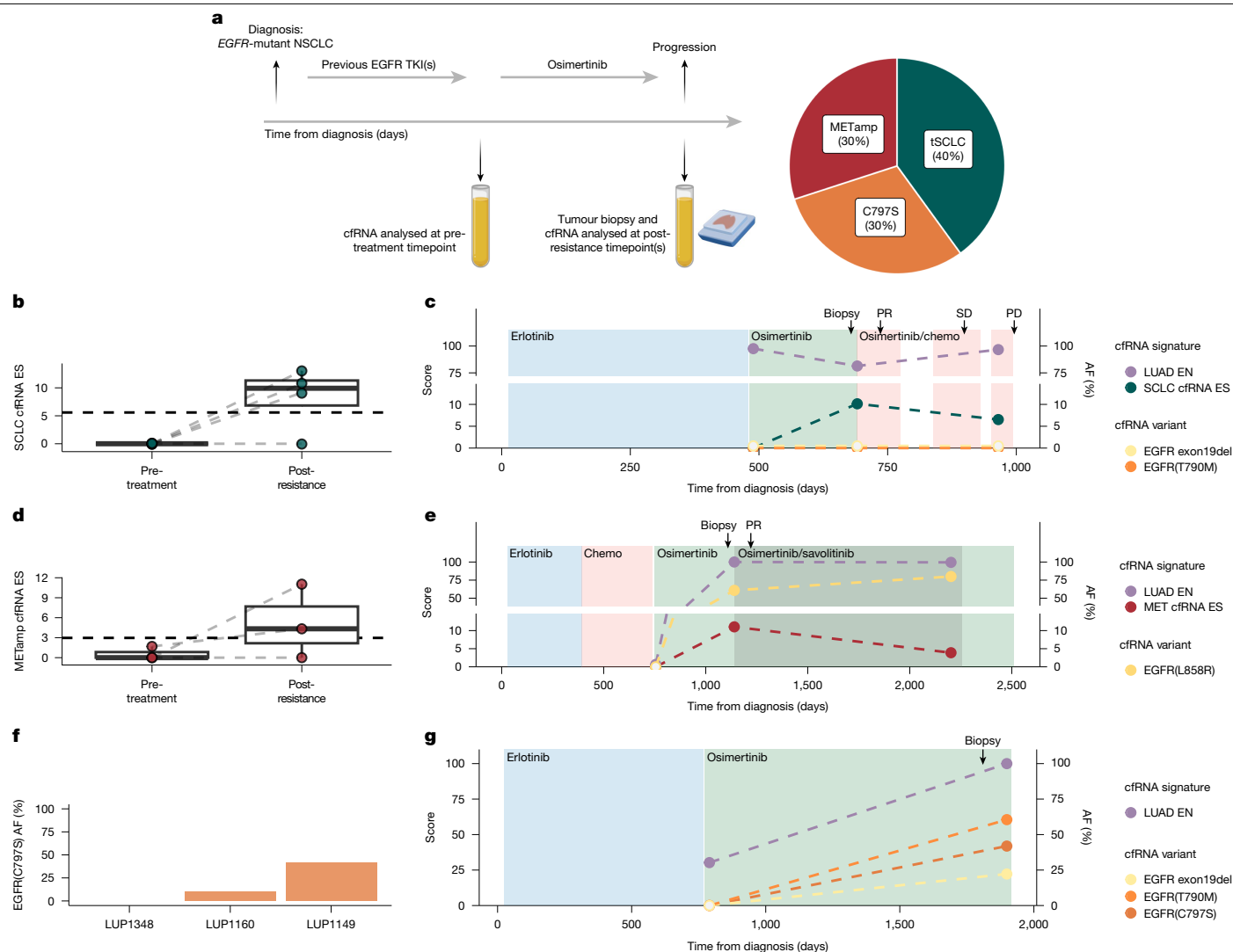


Fig. 4 | Detecting mechanisms of resistance to EGFR TKIs in cfRNA.

a, Summary of the EGFR TKI cohort ($n = 10$ patients, $n = 24$ timepoints). Mechanisms of resistance were determined via tissue biopsy. *C797S*, *EGFR*^{C797S} SNV; METamp, *MET* amplification; tSCLC, histological transformation to SCLC. The illustrations of the plasma tubes and tissue block were created using BioRender (<https://biorender.com>). **b**, Enrichment scores for SCLC Sig ($n = 73$ genes) in patients with biopsy-proven tSCLC. The threshold representing 95% or more specificity in controls is shown by the dashed line. **c**, Vignette of a patient with tSCLC after EGFR TKI treatment. The left axis depicts the enrichment or elastic net scores, and the right axis depicts the allele frequency (AF) of

tumour-adjudicated variants detected in cfRNA. PD, progressive disease; PR, partial response; SD, stable disease. **d**, Enrichment score for a METamp Sig ($n = 9$ genes) in patients with biopsy-proven *MET* amplification. The threshold representing 95% or more specificity in controls is shown by the dashed line. **e**, Vignette of a patient with emergence of *MET* amplification after EGFR TKI treatment. **f**, *EGFR*^{C797S} allele frequency in cfRNA from patients with biopsy-proven *C797S* mutation. **g**, Vignette of a patient with emergence of a *EGFR*^{C797S} mutation after EGFR TKI treatment. In the boxplots, the centre line denotes the median, the box contains the IQR, and the whiskers denote extrema no farther than $1.5 \times$ IQR.

(AUROC = 0.95; Fig. 3l,m). Performance of the elastic net classifier was statistically similar to the enrichment score framework in the training cohort ($P = 0.27$), but improved performance in the validation cohort ($P = 0.001$). Therefore, when sufficient samples are available, machine learning-based approaches can also be used to train robust classifiers for RARE-seq.

Identifying resistance to EGFR therapies

One of our motivations for developing RARE-seq was the hypothesis that in addition to identifying emergent somatic mutations mediating resistance, ctRNA analysis might allow detection of non-genetic therapeutic resistance mechanisms, such as histological transformation¹⁹. To explore this question, we focused on patients with *EGFR*-mutant LUAD treated with EGFR tyrosine kinase inhibitors (TKIs). Acquired resistance to EGFR TKIs is caused by heterogenous mechanisms, including both

EGFR-dependent mechanisms (such as secondary point mutations in the *EGFR* gene) and EGFR-independent mechanisms (such as bypass pathway activation or histological transformation)^{20,21}. We profiled plasma samples from 10 patients with stage IV *EGFR*-mutant LUAD who had received osimertinib after progressing on at least one previous EGFR TKI and who then developed resistance to osimertinib. The mechanism of resistance was determined by tissue biopsy at the time of progression, revealing small-cell histological transformation in four patients, *MET* gene amplifications in three patients, and emergent *C797S* point mutations in *EGFR* in another three patients (Fig. 4a). For each patient, blood was collected before the start of osimertinib and at one or more timepoints after progression ($n = 24$ timepoints total).

Histological transformation to small-cell lung cancer (SCLC) was evaluated using a SCLC signature (SCLC Sig) defined using public tumour gene expression data^{18,22}. The signature comprised 73 genes,

including canonical SCLC markers such as *ASCL1*, *NEUROD1* and *INSM1*. At time of radiological progression, the SCLC Sig was detected in three of four (75%) patients with biopsy-proven histological transformation (Fig. 4b). For one of these patients, we also analysed a plasma sample collected 274 days after the initiation of SCLC-targeted chemotherapy (for example, carboplatin/etoposide) and observed decreased SCLC cfRNA enrichment score, suggesting treatment response of the small cell component (Fig. 4c).

We next assessed the performance of RARE-seq for identifying genetic mechanisms of resistance. To detect MET pathway activation caused by *MET* amplification using ctRNA, we developed a *MET* amplification signature (METamp Sig) containing nine genes differentially expressed in three *EGFR*-mutant NSCLC cell lines with acquired resistance due to *MET* amplification (Methods). The METamp Sig was detected in two of three (67%) patients with biopsy-proven *MET* amplification (Fig. 4d). Of note, we observed a decreased METamp cfRNA enrichment score in one patient after the MET inhibitor savolitinib was added to osimertinib, coincident with a partial response by imaging (Fig. 4e). We then identified *EGFR*^{C797S}-mutant reads in two of three (67%) patients known to have acquired this mutation based on tumour DNA or ctDNA analysis (Fig. 4f,g). These resistance mechanisms were not detected in any pre-resistance timepoint, confirming the specificity of our approach (Supplementary Table 6). Together, these results demonstrate proof of concept that RARE-seq can identify both non-genetic and genetic mechanisms underlying *EGFR* TKI resistance, including histological transformation, which is not detectable by mutation-based ctDNA approaches.

Tissue-of-origin determination in ctRNA

We also wanted to test whether the same RARE-seq assay could be used to detect other cancer types. We therefore analysed plasma samples from patients with three types of other carcinomas: pancreatic adenocarcinoma (PAAD; $n = 10$ patients with stage IV), prostate adenocarcinoma (PRAD; $n = 1$ patient with stage III and $n = 8$ patients with stage IV) and liver hepatocellular carcinoma (LIHC; $n = 3$ patients with stage I, $n = 5$ patients with stage II and $n = 2$ patients with stage IV). Cancer-specific gene signatures were generated for each cancer type and evaluated using the cfRNA enrichment score framework (Supplementary Table 7). Sensitivity of cancer-specific cfRNA enrichment score detection was 80%, 90% and 100% for LIHC, PAAD and PRAD, respectively, after adjusting for multiple hypothesis testing (Fig. 5a). Given that cancer-specific signatures were detectable in multiple cancer types, we next wanted to explore whether ctRNA analysis can distinguish between cancer types. The ability to identify the TOO of a malignancy non-invasively could be helpful for patients with advanced-stage cancers of unknown primary who have no clinically detected primary tumour. Thus, we generated additional cancer signatures capturing genes uniquely expressed in each cancer compared with all other cancers, as well as to cfRNA from patients without cancer (cancer-TOO signatures). TOO was assessed for any advanced-stage cancer sample detected by at least one cancer-specific signature. The top predicted TOO was correct 89% of the time and the top two predicted TOOs identified the correct tumour type 96% of the time (Fig. 5b,c). Thus, cfRNA analysis could potentially enable non-invasive TOO identification.

Application to non-malignant conditions

We also explored several potential applications of RARE-seq beyond oncological settings, by first testing whether cfRNA expression might reveal lung injury caused by benign pulmonary conditions. We profiled cfRNA from 3 patients with chronic obstructive pulmonary disorder, 7 patients with active COVID-19 infections ($n = 13$ timepoints) and 19 patients admitted to the intensive care unit with evidence of

pulmonary injury (including 9 patients with acute respiratory distress syndrome). Across these diverse pulmonary conditions, lung-derived cfRNA was detectable in 49% of patients using a normal lung gene expression signature derived from GTEx¹⁶ (Fig. 5d and Supplementary Table 8). By contrast, among adult participants without known pulmonary conditions ($n = 42$), lung cfRNA was detected in 14%, with detection being significantly more frequent in current smokers than in former- or never-smokers (45.5% versus 3.6%, $P = 0.004$; Fig. 5e). This result suggests that lung cfRNA is induced by ongoing injury to pulmonary epithelium caused by recent exposure to tobacco smoke^{23,24}. Conversely, in patients with known pulmonary conditions, detection was significantly greater in those who were on a ventilator at the time of blood collection (Fig. 5f), probably reflecting the severity of the pulmonary condition as well as insult to pulmonary epithelium from mechanical ventilation²⁵. Thus, cfRNA analysis may have utility for assessment of benign conditions involving acute and chronic patterns of tissue injury.

Separately, given the rapidly growing interest in RNA-based vaccines and therapeutics, we explored whether RARE-seq might have utility for simultaneously tracking RNA-based treatments and their effects on the host. To do so, we longitudinally profiled plasma from five individuals receiving mRNA-based COVID-19 vaccinations ($n = 20$ timepoints). We performed RARE-seq on these samples using the RAG capture panel supplemented with baits targeting the sequence of the mRNA vaccine (Extended Data Fig. 3c). Vaccine RNA was detectable immediately after vaccination and persisted for up to 6 weeks after injection (Fig. 5g). Host genes that were differentially expressed after vaccination were enriched for inflammatory responses, interferon- γ responses and cytokine signalling, reflecting immune activation in response to vaccination (Fig. 5h). Vaccine RNA and host immune response genes were both highest in the first week post-vaccination (Fig. 5i). These results highlight the potential of RARE-seq for simultaneously measuring pharmacokinetic and pharmacodynamic aspects of RNA-based therapies.

Discussion

Here we describe a method for cfRNA profiling, demonstrate its performance in NSCLC and provide proof-of-concept results for various additional clinical applications. After end-to-end optimization, we found that RARE-seq achieves a limit of detection below that of tumour-naïve ctDNA-based methods^{26–30}, resulting in a higher sensitivity for detection of NSCLC. A key innovation of our approach is the specific capture of transcripts that are absent or very lowly expressed in plasma of healthy individuals, facilitating improved sensitivity for transcripts stemming from non-haematological tissues. These transcripts are highly enriched for cancer-specific and solid organ-specific genes and therefore useful for detection of pathophysiology occurring outside of the blood. RAG-focused RARE-seq can be applied to diverse pathophysiological conditions without the need for bespoke capture panels.

One important observation was the high proportion of platelet-derived transcripts present in cfRNA. In contrast with cfDNA shed by megakaryocytes³¹, these transcripts largely stem from intact platelets that partition with plasma during blood sample processing and represent a major confounder for cfRNA analyses. Indeed, expression differences in cfRNA of patients with cancer versus controls reported in several previous studies appear to largely stem from differences in platelet-derived transcripts^{2,32,33}. We therefore developed a computational approach for eliminating confounding effects caused by platelet RNA and found that this significantly improved sensitivity for detection of lung cancer. Although platelet contamination can also be reduced using pre-analytical approaches such as increasing centrifugation speed³⁴, even platelet poor plasma preparations are seldom entirely platelet free. In addition, such procedures are not feasible when analysing previously banked samples. Therefore, we anticipate that our

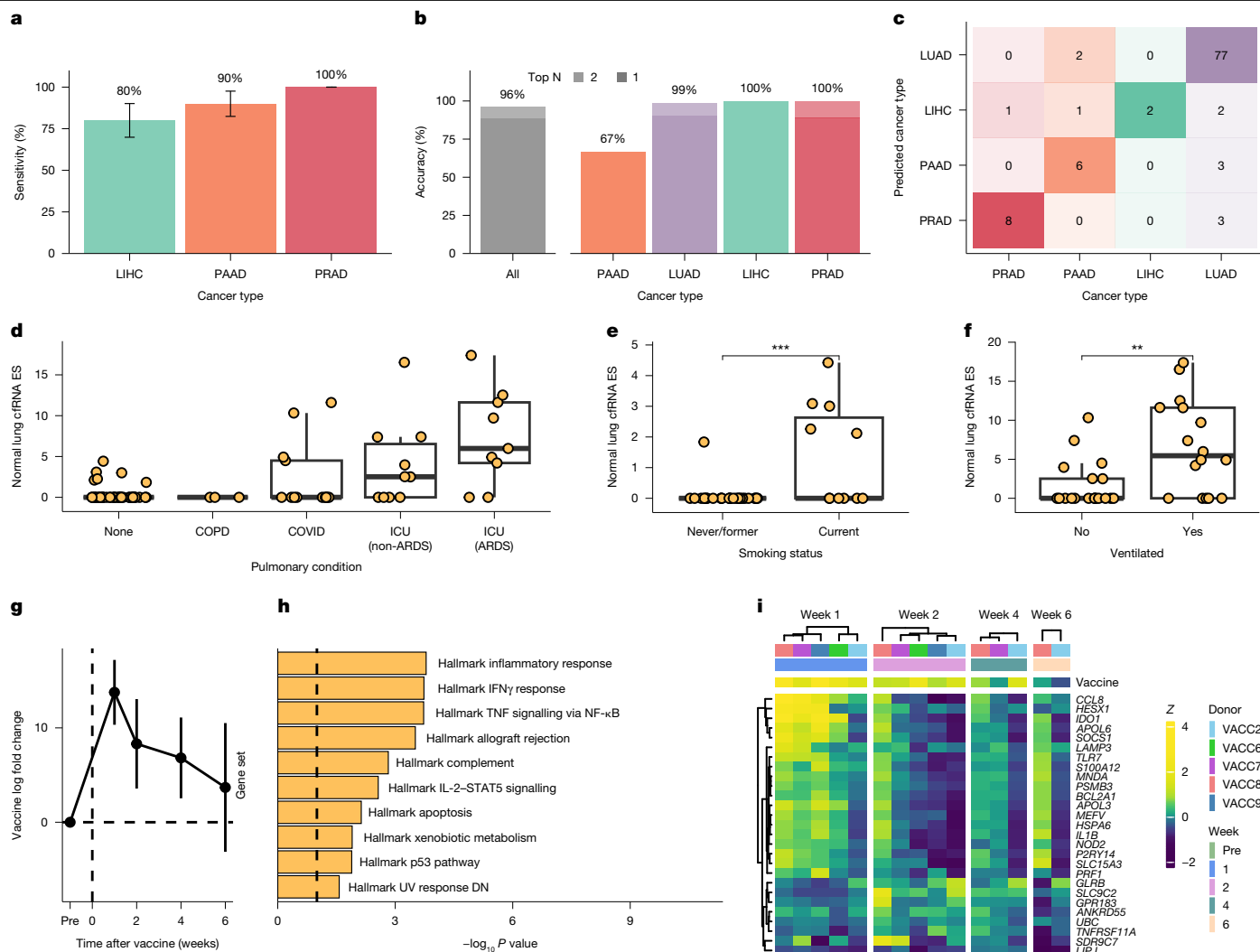


Fig. 5 | Exploring additional potential applications of RARE-seq.

a, Sensitivity of detection of LIHC cRNA ($n = 10$), PAAD ($n = 10$) and PRAD ($n = 9$) enrichment scores using the respective cancer-specific signatures. Detection thresholds achieving 90% or more specificity after multiple hypothesis correction were used. Error bars represent the 95% binomial confidence interval. **b**, Accuracy of TOO determination using cancer-TOO signatures in samples detected by at least one cancer-specific signature ($n = 105$). The shading depicts enrichment score rank for the true cancer type. **c**, Confusion matrix comparing predicted and true cancer type. **d**, Enrichment score for normal lung tissue signature ($n = 5$ genes) in cRNA from patients with benign pulmonary conditions ($n = 77$ timepoints from 71 individuals). ARDS, acute respiratory distress syndrome; COPD, chronic obstructive pulmonary disease; COVID, COVID-19 infection; ICU, intensive care unit. **e**, Relationship between normal lung enrichment score and smoking history. Samples without known

pulmonary conditions were compared using two-sided Wilcoxon test ($n = 42$, $P = 0.00095$). **f**, Relationship between normal lung enrichment score and ventilator status. Samples with known pulmonary conditions were compared using two-sided Wilcoxon test ($n = 35$ timepoints from 29 individuals, $P = 0.0061$). **g**, Timecourse depicting COVID-19 mRNA vaccine expression in cRNA ($n = 20$ timepoints from 5 individuals). Data are presented as mean log fold change above baseline $\pm$ s.d. **h**, GSEA of DEGs in post-vaccination timepoints. DEGs were determined by a likelihood ratio test, and GSEA significance was determined using fgsea (Methods). The top ten MSigDb Hallmark sets are displayed^{37–39}. DN, down. **i**, Heatmap showing COVID-19 mRNA vaccine and DEGs in post-vaccination timepoints. Expression was centred and scaled using pre-vaccination levels. In the boxplots, the centre line denotes the median, the box contains the IQR, and the whiskers denote extrema no farther than $1.5 \times$ IQR.

approach for removing platelet contributions to cRNA gene expression profiles will be broadly useful for future cRNA studies.

Our exploration of potential applications suggests that ultrasensitive cRNA analysis may be useful in diverse clinical settings. For example, a promising application is non-invasive identification of cancer types. Such an approach could be useful for non-invasive identification of TOO in patients with cancers of unknown primary or distinguishing between cancer subtypes. As our approach enables tumour-naïve detection of ctRNA, it could potentially be used for cancer screening, either alone or in combination with other diagnostic modalities. However, future studies evaluating large clinical cohorts of patients with early-stage cancers and risk-matched controls will be required to rigorously test cRNA for screening applications. The fact that RARE-seq detected lung cancer

ctRNA in many samples that were negative by tumour-naïve ctDNA analysis suggests that the two approaches may be complementary.

When considering patients with *EGFR*-mutant NSCLC treated with *EGFR* TKIs, the ability of RARE-seq to simultaneously query for the presence of somatic mutations (for example, *EGFR*^{C797S}) and the transcriptional effects of pathway activation via somatic alterations (for example, *MET* amplification) or non-genetic mechanisms (for example, histological transformation) allows more comprehensive non-invasive profiling of resistance mechanisms than previously possible. As histological transformation is a common mechanism of resistance that occurs in diverse cancer types³⁵ and because its diagnosis currently requires invasive tissue biopsies, we envision that its detection using cRNA will be of broad interest. Future studies focused on

each of the potential clinical applications that we explored here will be required to validate our findings and to fully establish the utility of cfRNA analysis in these settings. In addition, it will be interesting to compare and potentially combine RARE-seq with methylation-based and fragmentomics-based liquid biopsy approaches.

Finally, we envision that RARE-seq will be useful in detecting non-cancer-derived cfRNA signatures in patients with both malignant and benign conditions. For example, our data demonstrating the presence of normal lung RNA in the plasma of patients with acute lung injury due to conditions such as COVID-19 infection or acute respiratory distress syndrome suggest that cfRNA analysis may also allow blood-based monitoring of tissue injury. Separately, as demonstrated in the analysis of cfRNA in plasma samples after COVID-19 mRNA vaccination, we envision that RARE-seq will enable simultaneous tracking of mRNA vaccines or RNA therapeutics and measurement of host responses.

In summary, we developed a versatile method for cfRNA analysis that has potential utility in diverse clinical settings. Our results also include several important biological and technical insights that enable more sensitive cfRNA analysis and that will probably be broadly useful for cfRNA-based liquid biopsy applications. We anticipate that our approach could enable novel non-invasive diagnostic applications, with the promise of advancing precision medicine and improving patient care.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-08834-1>.

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Methods

Human participants and cohorts

All samples analysed in this study were collected with informed consent using protocols approved by Institutional Review Boards at their respective centres. Collection centres included Stanford University, Memorial Sloan Kettering Cancer Center (MSKCC), Massachusetts General Hospital (MGH) and University Hospital Zurich, as detailed below. In total, 459 blood samples were collected from 375 individuals (Extended Data Fig. 6). The clinical and demographic characteristics are presented in Extended Data Table 1 for cohorts without cancer and in Extended Data Table 2 for cohorts with cancer.

Cohorts with cancer. Blood samples were collected from patients with NSCLC enrolled at Stanford University ($n = 99$), MSKCC ($n = 40$) and MGH ($n = 52$). Of these, 138 (99%) were diagnosed with LUAD and 2 patients (1%) had mixed adenosquamous histology. Stage information was determined using the American Joint Committee on Cancer (AJCC) 8th edition. For 10 patients with *EGFR*-mutant NSCLC enrolled at the MGH, blood was collected at multiple timepoints before treatment with and after progression on the *EGFR* TKI osimertinib, where available ($n = 24$ plasma samples). Tissue biopsies were collected at the time of progression to identify the mechanism of resistance. For additional exploratory analyses, blood samples were collected at Stanford University from patients diagnosed with advanced-stage PAAD ($n = 10$) and advanced-stage PRAD ($n = 9$), and at University Hospital Zurich from patients with LIHC ($n = 10$).

Cohorts without cancer. Blood from individuals with no known cancer or benign pulmonary conditions was collected at Stanford University for technical experiments and for method development controls ($n = 45$). A subset of these samples comprised the 'meta-reference' set that was used to define expected expression in healthy cfRNA for various downstream analyses ($n = 15$). Blood samples were collected from individuals at the time of LDCT screening at Stanford University ($n = 27$) and at the MGH ($n = 50$), who qualified for such screening based on smoking history (30-pack years or more) and age (55–80 years). Three individuals at the time of LDCT screening were noted to have chronic obstructive pulmonary disease. In addition, blood samples were collected from patients who were critically ill admitted to the intensive care unit at Stanford Hospital ($n = 20$). A subset of patients met rigorous definition of acute respiratory distress syndrome at the time of blood draw ($n = 9$). Finally, blood samples were collected from individuals at Stanford University with coronavirus disease 2019 (COVID-19; $n = 13$ timepoints from $n = 7$ individuals) as well as from individuals after receiving COVID-19 mRNA vaccination (Pfizer-BioNTech or Moderna; $n = 20$ timepoints from $n = 5$ individuals).

Blood collection and plasma processing

Peripheral blood samples were collected and processed according to protocols at their respective centres; sample processing details are provided in Supplementary Table 9. For technical experiments conducted at Stanford University, whole blood was collected in EDTA tubes and plasma isolated using centrifugation at 2,500g for 10 min at 4 °C. Optical density at 414 nm was measured using a Nanodrop spectrophotometer instrument to quantify levels of haemolysis in plasma⁴⁰. After centrifugation, all plasma was stored at –80 °C until cell-free nucleic acid isolation.

cfRNA extraction

Cell-free nucleic acids were extracted from plasma using the miRNA protocol from the QIAamp Circulating Nucleic Acid kit (Qiagen; range 0.2–8.0 ml). Extraction was performed according to the manufacturer's instructions with minor modifications. The resulting eluate was

incubated with 14 U DNase I (RNase-Free DNase Set, Qiagen) for 30 min at room temperature to digest DNA. The digested eluate was purified using the Zymo RNA Clean&Concentrator kit and stored at –80 °C.

Plasma processed before February 2021 was extracted using phenol/chloroform phase separation and purified using the QIAamp Viral RNA kit (Qiagen). In brief, plasma was first incubated with 3 volumes of TRI Reagent LS (Molecular Research Center), and then incubated with 0.4 volumes of chloroform (relative to plasma input). Phase separation was performed using Maxtract 50-ml conical tubes (Qiagen) spun at 1,500g for 5 min at room temperature. The RNA-containing aqueous phase was carefully removed, and RNA was extracted according to the QIAamp Viral RNA kit protocol. DNA digestion and clean-up were performed as described above. For the subset of samples digested using the 'on column' protocol, 28 U DNase I was added directly to the sample bound to the silica column and incubated for 15 min, according to the manufacturer's recommendations. Careful analysis was done to compare cfRNA yield (Extended Data Fig. 1d), gene expression distribution (Extended Data Fig. 2b) and LUAD cfRNA ES detection rates (Extended Data Fig. 4g) between different extraction methods. As no significant differences were found, cfRNA samples extracted using both methods were combined for the analyses presented in this article.

cfRNA quantification

Quantitative real-time PCR (RT-qPCR) was used for quantification of cfRNA. An RNA-specific primer was designed to cover a 97-bp amplicon spanning 2 exon boundaries in the housekeeping gene *GAPDH* (forward 5'-GATCATCAGCAATGCCTCCT-3', reverse 5'-TGTGGTCATGAGTCCTTCCA-3'). A DNA-specific primer was designed to cover a 78-bp transcriptionally silent region of chromosome 12 (forward 5'-TACGGTGGTCCTTTCTTCG-3', reverse 5'-TTTCCTTTGGGTCTGAATGC-3'). Reverse transcription was first performed using the High-Capacity cDNA Reverse Transcription kit (Applied Biosystems). qPCR was then run using 2X Power SYBR Green PCR Master Mix (Thermo Fisher Scientific) on an Applied Biosystems 7500 Fast Real-Time PCR or QuantStudio 7 Pro instruments. Universal Human Reference RNA (Thermo Fisher Scientific) was run in parallel to generate a standard curve, and cfRNA concentrations were calculated by comparing the RNA-specific Ct value of the sample to the standard curve. If DNA was detected using the DNA-specific primer, DNA digestion, clean-up and quantification were repeated. The cfRNA size distribution was assessed using the Agilent Bioanalyzer RNA 6000 Pico chip.

cfRNA library preparation and sequencing

RARE-seq. We targeted an input mass of 424 pg cfRNA, which represents the 25th percentile of cfRNA yield for 4 ml of plasma in healthy controls (Extended Data Fig. 1b). For samples with less than 424 pg, all extracted cfRNA was used for library preparation (range 2–424 pg). Double-stranded complementary DNA (cDNA) was synthesized from cfRNA using the NEBNext Ultra II RNA First-Strand Synthesis Module and Non-Directional Second Strand Synthesis Module (New England Biolabs). Double-stranded cDNA was treated with 100 U S1 nuclease (Thermo Fisher) for 30 min at room temperature to hydrolyse incomplete (that is, single stranded) regions. The KAPA Hyper Prep kit (Kapa Biosystems) was used to prepare libraries for sequencing, following the manufacturer's instructions with slight modifications as previously described³⁶. Whole-coding transcriptome capture was performed using the Roche Nimblegen SeqCap EZ MedExome Target Enrichment Kit and or the Twist Biosciences Comprehensive Exome Hybridization kit, following the respective manufacturer's instructions. Single-plex capture with a custom gene panel targeting rare abundance genes (see 'RAG-focused capture panel design' below) was performed using the Twist Biosciences Hybridization kit. Captured libraries were sequenced using 2 × 150-bp paired-end reads on Illumina HiSeq4000 or NovaSeq6000 instruments.

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SMART-seq. cRNA from three controls was input into library preparation using the SMART-Seq Stranded kit (TaKaRa) per the manufacturer's instructions and without additional fragmentation. Libraries were amplified and sequenced using 150-bp paired-end runs on an Illumina NovaSeq6000.

Leukocyte RNA processing

After plasma separation, leukocytes were isolated from plasma-depleted whole-blood samples using SepMate PBMC isolation tubes (StemCell Technologies), according to the manufacturer's instructions. Leukocyte cell pellets were stored at -80°C . RNA was extracted from leukocyte cell pellets by mixing with 800 μl TRIzol (Invitrogen) and 200 μl chloroform, followed by centrifugation at 13,000g for 15 min at 4°C . RNA was then purified from the aqueous supernatant using the Qiagen RNeasy kit, per the manufacturer's instructions. Total RNA was quantified with the NanoDrop instrument and Agilent Bioanalyzer RNA 6000 Pico chip. RNA (5–10 ng) was used for RARE-seq library preparation and whole-coding transcriptome capture as previously described. Fragmentation was performed before first strand synthesis as recommended by the manufacturer.

NCI-H1975 cell line RNA processing

NCI-H1975 LUAD cells were acquired from the American Type Culture Collection (ATCC CRL-5908). Cellular RNA was extracted from cell pellets and quantified as described for leukocyte RNA. To create in vitro NCI-H1975 cell line spikes, NCI-H1975 RNA was serially diluted into cRNA from a single healthy individual to create samples with 10%, 1%, 0.1%, 0.01% and 0.001% cancer fraction by mass. Triplicate RARE-seq libraries were generated from each mixture as well as NCI-H1975 RNA (100% cancer fraction) and cRNA alone (0% cancer fraction) and captured with whole-coding transcriptome panel. Additional mixtures using cRNA from a different healthy individual were created for capture with the RAG panel (see 'RAG-focused capture panel design' below). Cell line authentication was performed by confirming the presence of validated *EGFR*, *TP53*, *PIK3CA* and *CDKN2A* variants in RNA-seq data, and tissue culture facilities were regularly tested for mycoplasma contamination.

cfDNA library preparation and sequencing

cfDNA was extracted from plasma using the standard cfDNA protocol from the QIAamp Circulating Nucleic Acid kit (Qiagen). After extraction, cfDNA was quantified with the Qubit double-stranded DNA High Sensitivity kit (Thermo Fisher Scientific) and High Sensitivity NGS Fragment Analyzer (Agilent). Cancer personalized profiling by deep sequencing (CAPP-seq) was used to create sequencing libraries from 32 ng cfDNA, as previously described³⁶. Thereafter, hybridization-based capture (Roche NimbleGen) was used to target genes recurrently mutated in lung cancer³⁶. Libraries were sequenced using 2×150 -bp paired-end reads on Illumina HiSeq4000 instruments.

Mapping, deduplication and quality control for RARE-seq

FASTQ files were first demultiplexed using a custom pipeline, as previously described³⁶. Fastp (v0.20.0) trimmed the first 10 bases from the 5' end of read 1 and the 3' end of read 2, as well as removed low-quality reads (that is, reads with more than 20% of bases with Phred score < 30) or fragments shorter than 35 bp from each sample. Remaining high-quality reads were aligned to the reference transcriptome (GENCODE v27) and to the human genome (hg19) using STAR 2-pass⁴¹ (v2.7.0). PCR duplicates were removed from both transcriptome-aligned and genome-aligned files using a custom barcoding approach⁴² (samtools v1.11). Deduplicated reads were used for gene-level expression estimation using RSEM⁴³ (v1.2.28).

Quality control was assessed using metrics calculated by the RNASeQC package (v2.3.5) such as read mapping quality and mapping rates, including exonic, intronic and intergenic rates, and ribosomal

RNA rates. In addition, DNA contamination was estimated by calculating the percent of reads that map to intronic sequences out of the total number of reads that span an exon boundary. Samples were removed from downstream analysis if they had fewer than 20 million reads or if they had greater than 20% estimated DNA contamination. Together, these quality control thresholds removed four samples (Supplementary Table 9).

Gene expression normalization and platelet correction

RSEM expected counts for captured genes were used for expression analyses (tximport R package v1.22). Counts were first normalized using the trimmed mean of M values (TMM) method, which accounts for sample-to-sample variation in library size and transcriptome complexity (edgeR R package v3.36). Log-transformed and normalized expression values are referred to as ' $\log_2\text{NX}$ ' in this article. A modified version of the remove unwanted variation (RUV) approach⁴⁴ was then used to correct for expression variation caused by platelets (RUVseq R package v1.28). In brief, this method estimates factors of unwanted variation in gene expression using negative control genes (RUVg) or negative control samples (RUVs) that are impacted by unwanted technical or biological noise, but not the biological covariates of interest (for example, cancer). For whole-coding transcriptome libraries, the factor of unwanted platelet variation was estimated by RUVg using 130 platelet cell-type marker genes from PanglaoDB¹² as the set of negative control genes (Fig. 1i). For RAG-captured libraries, as most platelet genes were intentionally excluded from the panel (see 'RAG-focused capture panel design' below), factors of unwanted variation were estimated using the healthy control meta-reference cRNA ($n = 15$) as negative control samples within RUVs. The factor of unwanted platelet variation was selected to be the factor that most correlated with platelet expression, defined as average $\log_2\text{NX}$ across 21 captured platelet cell-type marker genes (for example, *PPBP*, *F5*, *F7*, among others; Extended Data Fig. 3g). In both cases, platelet-corrected gene expression was then determined using the residuals from ordinary least squares regression on the factor of unwanted platelet variation. RUVg and RUVs performed similarly in whole-coding transcriptome samples (Fig. 1j and Extended Data Fig. 3h). After normalization and correction, quality control was performed by calculating the Pearson correlation between the gene expression profile of each sample and the average of all other samples from the same sample group (that is, 'within-group correlation'). For controls with more than one biological replicate, the replicate with the highest within-group correlation was selected for downstream analyses.

Differential gene expression analysis

Differential gene expression analysis was performed using DESeq2 (DESeq2 R package v1.34)⁴⁵. For analysis of cRNA and cellular RNA differential expression, a generalized linear model (GLM) was built using sample type as the covariate. For analysis of cancer versus control differential expression, the GLM used condition (that is, cancer or control) and the factor of unwanted platelet variation determined by RUV as covariates. Wald test was used to determine significance of log fold change estimates followed by Benjamini–Hochberg multiple hypothesis correction. For the COVID-19 vaccination time series analysis, significance of differential expression was evaluated using a likelihood ratio test comparing the full GLM including vaccine timepoint and factor of unwanted variation compared with a reduced model with vaccine terms removed. For all analyses, significantly DEGs were defined by absolute log fold change greater than one and adjusted $P < 0.05$. Functional analysis of DEGs used GSEA of pre-ranked log fold change estimates⁴⁶ (fgsea R package v1.20). Significance was determined using an adaptive multilevel split Monte Carlo scheme and adjusted using Benjamini–Hochberg multiple hypothesis correction. GSEA was performed using either cell marker genes from the PanglaoDB single-cell RNA sequencing database¹² and/or the molecular signatures database (MSigDb v7.1)^{37–39}.

RAG-focused capture panel design

To identify RAGs in cfRNA, gene expression data were analysed from controls generated with the whole-coding transcriptome RARE-seq method ($n = 50$ biological replicates from $n = 28$ individuals). Expression was normalized using TMM followed by platelet correction, and genes were categorized as expressed or unexpressed using k -means clustering ($k = 2$) within each sample. RAGs were defined as genes that were expressed in less than 5% of cfRNA samples and for which average $\log_2 NX$ was in the bottom 30% of all genes. To further refine the RAG list, gene expression data from whole-blood samples generated by the GTEx project¹⁶ were accessed through the UCSC Xena repository⁴⁷ ($n = 307$). After TMM normalization and k -means clustering, RAGs were additionally defined as genes expressed in less than 40% of whole-blood samples. RAGs are listed in Supplementary Table 1. Next, expression uniformity was calculated in cfRNA using Gini coefficient, and endogenous control genes were selected from housekeeping genes⁴⁸ with a Gini < 0.2 in circulation. In addition, lung cancer-associated genes were manually curated for inclusion in the panel, including genes that are recurrently mutated or rearranged in lung cancer or that are aberrantly expressed in lung cancer histological types. In total, the RAG-focused capture panel included 80,866 probes targeting 7,766,820 bp and 5,546 unique genes, including 4,737 RAGs (Supplementary Table 2). Probes were also designed to target the sequences of the Pfizer-BioNTech and Moderna COVID-19 mRNA vaccines and were used to supplement the RAG capture panel for samples collected for vaccine time course analysis.

A subset of cfRNA libraries were captured using a second version (V2) of the RAG-focused capture panel. In total, the V2 RAG-focused capture panel included 96,360 probes targeting 9,256,091 bp and 6,847 unique genes (Supplementary Table 2); 5,079 (91.6%) genes and 4,594 (97.0%) RAGs were shared between panels. No significant differences in detection rate were observed between panels (Extended Data Fig. 4h), and so cfRNA samples captured using both panels were combined for the analyses presented in this article. Gene expression normalization and enrichment score detection were performed separately, using meta-reference controls captured with the respective panel. Differential expression analysis was performed using the full cohort but was restricted to the subset of shared genes between panels ($n = 5,079$ genes).

Cell type, tissue and cancer gene signatures

For cell-type-specific signatures, 7,481 marker genes from 170 human cell types were downloaded from the PanglaoDB single-cell RNA sequencing database¹².

Tissue-enriched and cancer-enriched signatures identified using gene expression data from the GTEx¹⁶ and TCGA projects were analysed. Gene-level read counts generated by the UCSC Toil RNA sequencing bioinformatic pipeline were accessed in the UCSC Xena repository⁴⁷. Gene expression was TMM normalized, and outlier samples were removed using the previously described within-group correlation metric. TCGA cohort was further filtered to exclude tumours with less than 60% tumour purity using previously reported consensus purity estimates⁴⁹. The normal tissue types that were evaluated from the GTEx included bladder ($n = 9$), brain ($n = 1,107$), breast ($n = 168$), colon ($n = 261$), oesophagus ($n = 598$), kidney ($n = 24$), liver ($n = 97$), lung ($n = 241$), ovary ($n = 79$), pancreas ($n = 145$), prostate ($n = 86$), skin ($n = 497$), stomach ($n = 167$) and whole blood ($n = 290$). Genes were defined as tissue enriched if expression was fivefold higher in each tissue than in all other tissues evaluated, and if the average tissue $\log_2 NX$ was greater than zero. Similarly, the cancer tissue types that were evaluated from TCGA included bladder (bladder urothelial carcinoma, BLCA, $n = 344$), brain (glioblastoma, GBM, $n = 144$; brain lower grade glioma, LGG, $n = 183$), breast (breast invasive carcinoma, BRCA, $n = 914$), colon (colon adenocarcinoma, COAD, $n = 270$), oesophagus (oesophageal

carcinoma, ESCA, $n = 181$), kidney (kidney chromophobe, KICH, $n = 65$; kidney renal clear cell carcinoma, KIRC, $n = 350$; kidney renal papillary cell carcinoma, KIRP, $n = 262$), liver (LIHC, $n = 164$), lung (LUAD, $n = 309$; lung squamous cell carcinoma, LUSC, $n = 365$), ovary (ovarian serous cystadenocarcinoma, OV, $n = 412$), pancreas (PAAD, $n = 177$), prostate (PRAD, $n = 473$), melanoma (skin cutaneous melanoma, SKCM, $n = 94$) and stomach (stomach adenocarcinoma, STAD, $n = 413$). Genes were defined as cancer enriched if expression was fivefold higher in each cancer tissue type than in all other cancer types evaluated, and if the average cancer tissue $\log_2 NX$ was greater than zero.

Cancer-specific gene signatures were created to include the genes most differentially expressed in each cancer tissue compared with whole blood and meta-reference cfRNA, and therefore most likely to be detected in cfRNA mixtures with high haematopoietic background. For this purpose, we supplemented the TCGA cancer cohort with gene expression data from the CCLE¹⁸ accessed through Xena (LIHC, $n = 25$; LUAD, $n = 76$; PAAD, $n = 40$, PRAD, $n = 7$ and SCLC, $n = 50$) and an additional SCLC study ($n = 79$)²². DESeq2 differential gene expression analysis quantified the gene-wise fold change between cancer tissue expression and whole blood and meta-reference cfRNA expression. Each gene was also categorized as expressed or unexpressed in cancer tissue, whole blood and meta-reference cfRNA using k -means clustering ($k = 2$) of average gene expression in each group. Genes were selected for cancer-specific signatures if they were: (1) found in the top 1% of genes overexpressed in the cancer tissue of interest or in whole blood and meta-reference cfRNA, and (2) categorized as expressed in the cancer tissue of interest or in whole blood and meta-reference cfRNA.

An analogous method was used to generate cancer TOO gene signatures except that gene expression was compared between LIHC, LUAD, PAAD, PRAD, whole blood and meta-reference cfRNA, rather than analysing each cancer type individually. Genes were selected for cancer TOO signatures if they were: (1) found in the top 1% of genes overexpressed in cancer tissue compared with whole blood and meta-reference cfRNA; (2) found in the top 5% of genes overexpressed in the cancer tissue of interest compared with each other cancer type; and (3) categorized as expressed in the cancer tissue of interest, and unexpressed in whole blood and meta-reference cfRNA.

For all signatures, a gene weight was defined as the sum of cancer tissue $\log_2 NX$ and cancer tissue versus meta-reference cfRNA log fold change, so that the weight preserved the sign of the differential expression analysis (that is, positive for genes overexpressed in cancer tissue). Cancer-specific and cancer TOO gene signatures and gene weights can be found in Supplementary Table 10.

Gene signature detection using cfRNA enrichment scores

The cfRNA enrichment score analytical framework was developed to be broadly applicable, as any gene signature of interest can be used. Gene expression of a sample was first standardized into Z scores using average and standard deviation of meta-reference cfRNA expression. Next, Z scores for a gene signature of interest were aggregated using the Stouffer equation and the gene weights described previously, resulting in a single enrichment score for each sample. If applicable, Z scores were aggregated separately for genes with positive and negative weights and summated. Bootstrapping was used to create random gene signatures ($n = 1,000$) of the same size (but excluding genes within the gene signature), and the bootstrapped enrichment score distribution was used to estimate an empirical P value for each signature in each sample. The enrichment score was set to zero if the empirical was $P < 0.05$. For analyses that considered more than one signature (that is, multi-cancer detection), P values were adjusted for multiple hypothesis testing using the Benjamini–Hochberg method.

Cancer TOO classification

Advanced-stage (III–IV) cancer samples that were detected by at least one cancer-specific gene signature were considered for TOO analysis.

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TOO was classified using a modified cfRNA enrichment score framework. Here Z scores were aggregated by calculating the 90th percentile of all Z scores in a gene signature of interest. For each sample, scores for all cancer TOO gene signatures were ranked. Accuracy was determined based on whether the diagnosed cancer type had the highest cancer TOO score detected, or the second-highest score.

LOD estimation using H1975 mixtures

The LOD95 of RARE-seq was determined using in silico RNA mixtures. Sequencing data for NCI-H1975 RNA and healthy control cfRNA were generated and analysed as described for the respective sample type. Reads were randomly subsampled from each replicate and merged in pre-specified cancer fractions ranging from 0% to 100%, and the cfRNA enrichment score analytical framework was used to detect cancer in each sample. Here an NCI-H1975 gene signature (H1975 Sig) was created by comparing NCI-H1975 samples processed by RARE-seq but not used for spike creation ($n = 4$) and meta-reference cfRNA ($n = 15$). Genes were selected if they were: (1) found in the top 1% of genes overexpressed in NCI-H1975 compared with meta-reference cfRNA, and (2) categorized as expressed in NCI-H1975.

The limit of blank was calculated from the average and standard deviation of H1975 cfRNA enrichment scores from additional cfRNA controls not used to create the gene signature ($n = 12$) and was set as the detection threshold for remaining spikes. Logistic regression was used to model the relationship between cancer fraction and H1975 Sig detection, and LOD95 was defined as the cancer fraction at which sensitivity exceeded 95%. This process was repeated for a range of RNA inputs and sequencing depths. The in vitro NCI-H1975 mixtures described above were processed as other cfRNA samples and detected using H1975 Sig to confirm the in silico results.

ctRNA variant calling

Variant calling was performed for a targeted list of clinically actionable hotspot alterations compiled based on the National Cancer Comprehensive Network guidelines for NSCLC⁵⁰. For candidate SNVs, the null hypothesis that the number of reads supporting the variant was consistent with a position-specific and depth-specific background error distribution was tested (Python v3.6.8). A beta-binomial model was used to estimate the background error distribution of each candidate SNV. The distribution was fit on a training cohort consisting of 50 control cfRNA samples (including 43 meta-reference cfRNA replicates) and used to evaluate the null hypothesis for each candidate test sample. Test samples included stage IV NSCLC samples with 1 or more tissue-adjudicated or ctDNA-adjudicated variants ($n = 136$) and healthy ($n = 28$) and LDCT ($n = 69$) controls. The resulting P values were adjusted for multiple hypothesis testing using the Benjamini–Hochberg method, with substitutions having $q < 0.1$ being considered significant. For a candidate SNV, if none of the training samples had a variant allele frequency > 0 , the SNV was automatically called for all test samples with allele frequency > 0 . Candidate fusions identified by STAR-Fusion⁴¹ (v1.6.0) were called if present in the list of known fusion partners in lung cancer as annotated in cBioPortal. *MET* exon 14 skipping alterations were called if the mapping contained splices linking exons 13–15 (samtools v1.12). Indels in *EGFR* exons 19 and 20 were called if greater than 6 bp. Variants and the respective ctRNA allele frequencies can be found in Supplementary Table 3.

ctDNA variant calling

CAPP-seq was performed as previously described³⁶ for the subset of patients with LUAD who had plasma available ($n = 38$). In brief, sequencing reads were demultiplexed, trimmed to remove low-quality bases from the 3' end using fastp, and aligned to the hg19 human genome using BWA ALN. PCR duplicates were removed using a custom barcoding approach. SNVs and indels were called in a tumour-naïve manner, as previously described⁵¹. Gene fusions were identified using FACTERA⁵².

Patient-specific variant lists were collected from previous clinical tumour genotyping and filtered to include only those variants that were also captured by the lung cancer CAPP-seq panel. The sample allele frequency was calculated by averaging the allele frequency of patient-specific, tumour-adjudicated variants that were also detected in ctDNA. ctDNA sample allele frequencies are provided in Supplementary Table 4.

Cancer detection using elastic net classification

Statistical learning models (that is, logistic regression-based elastic net) were trained to classify cfRNA from patients with stage IV LUAD ($n = 96$) from controls ($n = 47$; glmnet R package v4.1). Controls used for expression normalization or genotyping background distribution were excluded from this analysis. Features considered for model training included gene expression (that is, $\log_2$ NX), allele frequency from all recurrent, somatic variants queried in ctRNA, and binarized values representing whether SNV, fusions, indels, splice variants or variants of any type were detected. Pre-processing removed genes that (1) were not significantly differentially expressed between LUAD tumour tissue and meta-reference cfRNA (as described for the cfRNA enrichment score framework), (2) were differentially expressed in male versus female meta-reference controls, and (3) are located on sex chromosomes. In addition, both gene expression-based and mutation-based features with near-zero variance across samples were removed. Batch correction was performed to account for RAG capture panel (that is, V1 or V2) using the removeBatchEffect function (limma R package v3.50.3). Gene weights used for the cfRNA enrichment score framework were also evaluated as penalty weights within a weighted elastic net model⁵³, and model training was further restricted to only select features with non-negative coefficients. Nested tenfold cross-validation was used to tune hyperparameters and estimate model performance (caret R package v6.0). To evaluate the effect of cohort size on model performance, tenfold cross-validation was repeated after subsampling 20–100% of the cohort and the standard deviation of AUC was determined at each subsampling level across $n = 10$ random samples. The final weighted elastic net model selected 23 out of the 797 input features (Supplementary Table 10). The final model was validated using a withheld cfRNA cohort collected at an independent institution ($n = 52$ LUAD, $n = 46$ LDCT controls). ROC curves and metrics such as sensitivity, specificity and AUC were generated using the pROC R package (v1.18).

MET amplification gene signature

EGFR-mutant NSCLC cell line pairs with acquired resistance to *EGFR* TKI due to *MET* amplification (PC9–PC9-PERC17, MGH1157-1–MGH1157-3 and MGH170-1C #7–MGH170-1D #2) were generated. The PC9-PERC17 line was generated by in vitro culture of PC9 cells with erlotinib as previously described⁵⁴ and was a gift from S. Altschuler and L. Wu. MGH1157-1 and MGH1157-3 cell lines were developed from sequential biopsies of a patient before and after treatment with the combination of gefitinib and nazartanib, respectively. Clinical profiling of the MGH1157-3 tumour using *MET* FISH demonstrated *MET* amplification. MGH170 cell lines were developed from a patient with acquired *MET* amplification (confirmed by FISH) after erlotinib treatment⁵⁵. Single-cell clones were isolated from two separate metastatic lesions at the time of rapid autopsy following disease progression; the MGH170-1C #7 clonal cell line was sensitive to *EGFR* TKI, whereas the MGH170-1D #2 was found to be resistant. *MET* amplification was confirmed in all resistant cell lines, and *MET* dependency was confirmed by sensitivity to combination *EGFR* + *MET* TKIs. All tissue samples were obtained after patients signed informed consent to participate in a Dana-Farber–Harvard Cancer Center Institutional Review Board-approved protocol giving permission for research to be performed on their samples. mRNA-seq was performed on total RNA from pre-treatment and resistant cell lines ($n = 3$ replicates at each timepoint for $n = 3$ cell lines, or $n = 18$ samples total), and differential expression analysis was performed using DESeq2 with *MET* amplification status as covariate. Genes were

selected for a *MET* amplification gene signature if they were: (1) found in the top 1% of genes overexpressed in *MET*-amplified cell lines compared with meta-reference cfRNA; (2) significantly differentially expressed in *MET*-amplified cell lines compared with non-amplified cell lines; and (3) categorized as expressed in *MET*-amplified cell lines.

Here the gene weight was defined as the sum of the *MET*-amplified cell line $\log_2$ NX and *MET*-amplified versus meta-reference cfRNA log fold change, so that the weight preserved the sign of the differential expression analysis (that is, positive for genes overexpressed in *MET*-amplified cell lines). The gene signature is included in Supplementary Table 10.

Statistical analysis

All statistical analyses were performed in R (v3.6.0 and v4.1.3). Statistical tests used throughout the article include the Wilcoxon rank-sum test, paired Student's *t*-test, Fisher's exact test, Kruskal–Wallis analysis of variance, Pearson correlation, Spearman correlation, and DeLong's test for comparing AUC. Unless otherwise specified, statistical tests comparing two groups used the two-sided Wilcoxon rank-sum test. Unless otherwise specified, statistical tests comparing three or more groups used Kruskal–Wallis analysis of variance. Unless *P* values are stated, significance labels are used in figure panels as follows: NS, not significant, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$ and $****P < 0.0001$. Unless otherwise stated, multiple hypothesis correction was performed using the Benjamini–Hochberg method. Unless otherwise specified, the boxplots depict the median (centre line), interquartile range (box), and minimum and maximum (whiskers) that are no farther than 1.5× IQR (Tukey style). Unless otherwise specified, the sample size (*n*) provided in text, figure panels and figure legends refers to biologically independent individuals. Details of the statistical methods (including R packages) used in the cfRNA enrichment score analytical framework, elastic net model training and variant calling approaches are described in their respective section of Methods.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

De-identified clinical and demographic data for individuals considered in this study as well as gene signatures and associated enrichment scores, detected variants and other relevant data are provided in the Supplementary Tables. Gene expression data and data relevant for somatic variant calling that were generated during this study are available at <http://rareseq.stanford.edu>. Gene expression data from the GTEx¹⁶, TCGA¹⁷ and CCLE¹⁸ are publicly available from the UCSC Xena Repository⁴⁷ (<https://xenabrowser.net/datapages/>). Gene expression data from additional SCLC tumours²² can be accessed using the Gene Expression Omnibus (GSE60052). Cell-type marker gene lists are publicly available from the PanglaoDB resource¹² (<https://panglaoDB.se/markers.html>), and Molecular Signatures Database^{37–39} (MSigDb) gene lists are also publicly available (<https://www.gsea-msigdb.org/gsea/downloads.jsp>). Reference databases used in this study such as hg19 (https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.13) and GENCODE v27 (https://www.gencodegenes.org/human/release_27.html) can be accessed using the provided weblinks.

Code availability

The code generated during this study is available on GitHub (<https://github.com/RARESeq>).

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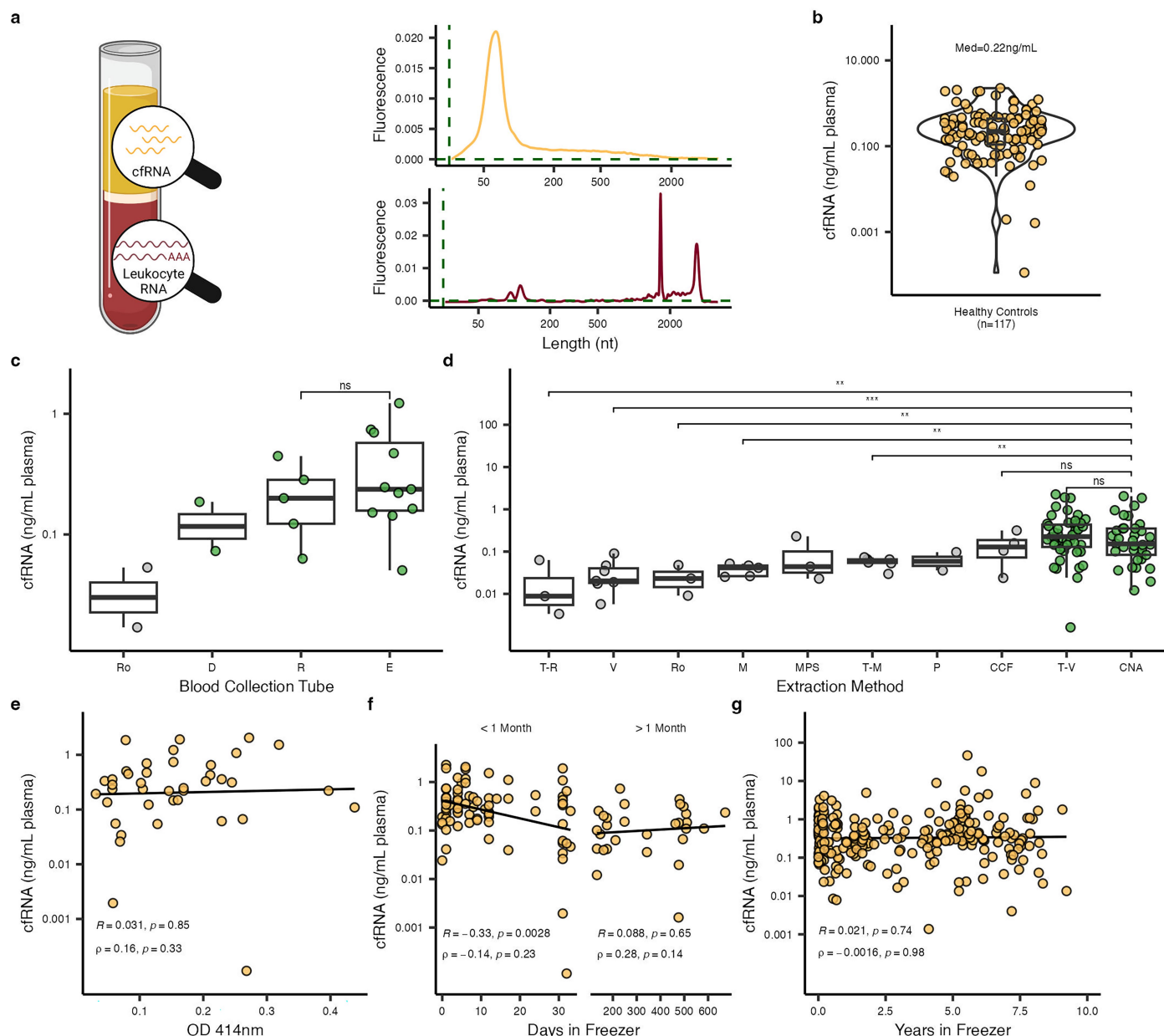
Additional information

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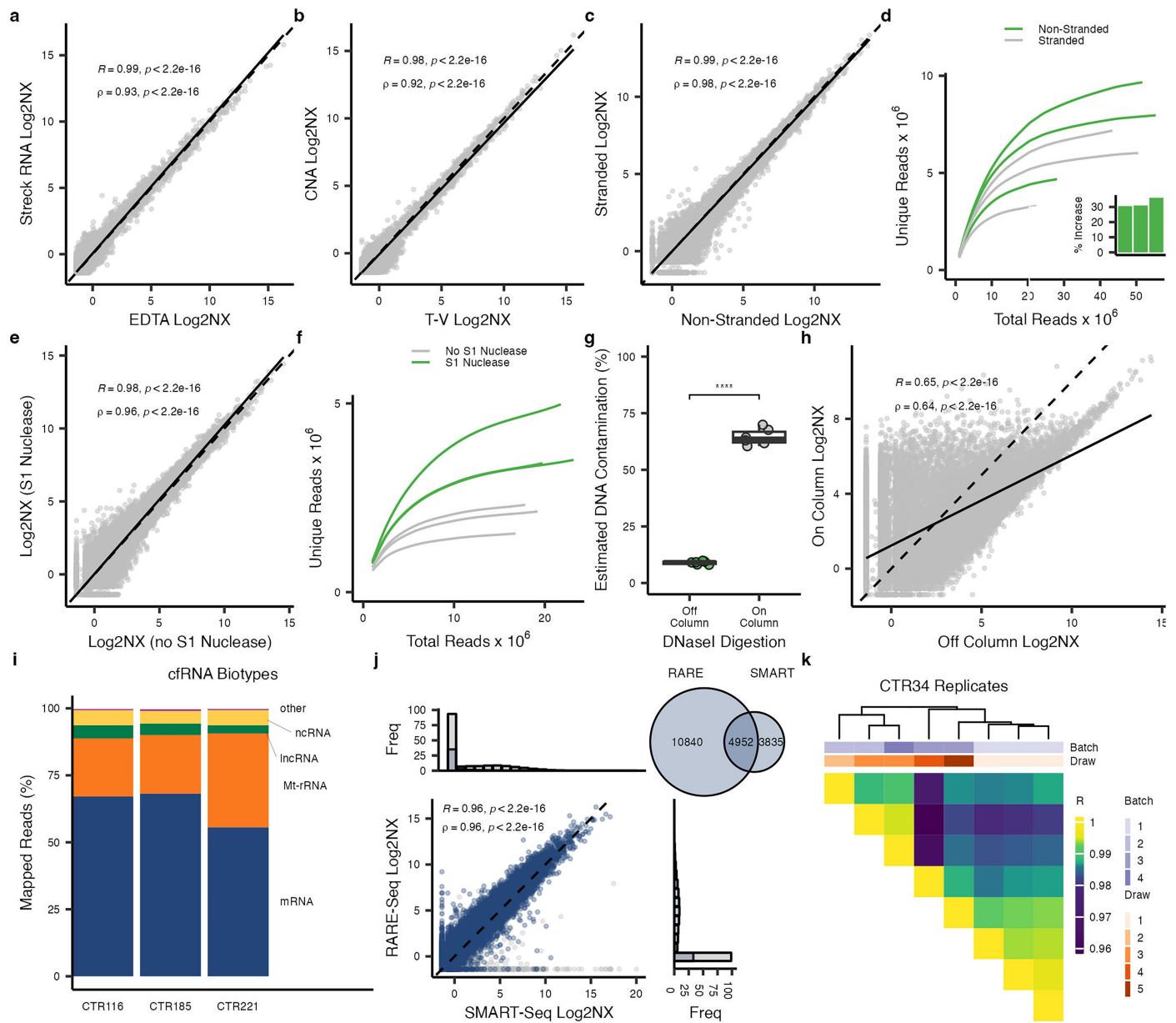
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Extended Data Fig. 1 | Optimization of blood collection and cfRNA

extraction. **a**, Representative bioanalyzer trace for cell-free RNA (yellow) and leukocyte cellular RNA (red). The blood tube was created using BioRender (<https://biorender.com>). **b**, Concentration of cfRNA per mL of plasma in healthy controls measured via quantitative PCR (see Methods; $n = 117$). Plasma was collected for technical experiments using 2,500 G centrifugation for 10 min. **c**, Analysis of plasma cfRNA concentrations using different blood collection tubes (BCTs). Ro, Roche Cell-free DNA BCT ($n = 2$). D, Streck Cell-free DNA BCT ($n = 2$). R, Streck RNA Complete BCT ($n = 5$). E, EDTA ($n = 11$). ns, not significant. **d**, Analysis of plasma cfRNA concentrations using different extraction methods. T-R, TRIzol-LS + Qiagen RNeasy Kit ($n = 3$). V, QIAamp Viral RNA Kit ($n = 7$). Ro, Roche High Pure Viral RNA Kit ($n = 3$). M, Qiagen miRNeasy Kit ($n = 5$). MPS, Qiagen miRNeasy Serum/Plasma Kit ($n = 3$). T-M, TRIzol-LS + Qiagen miRNeasy

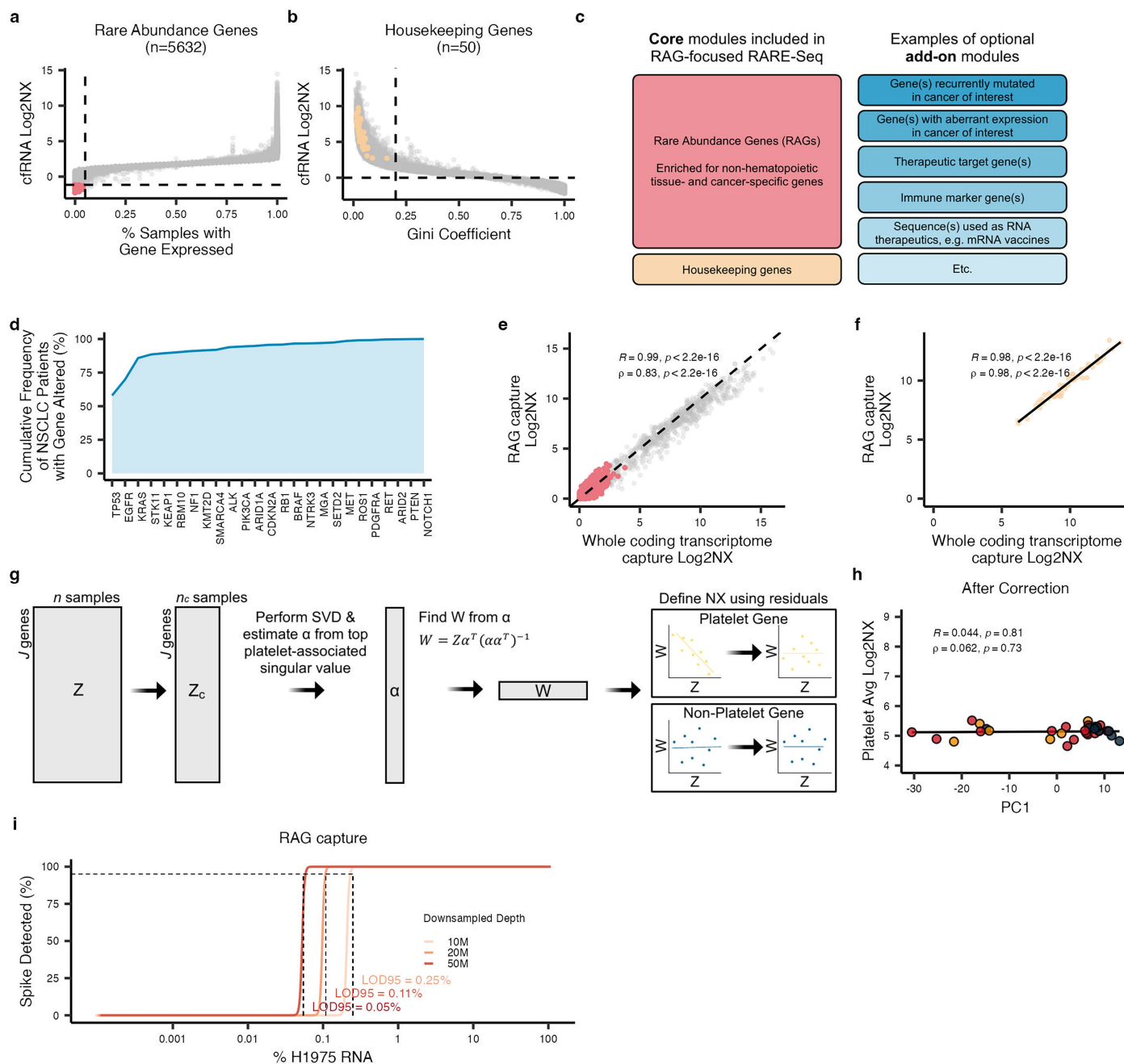
Kit ($n = 7$). P, mirVana PARIS Kit ($n = 2$). CCF, QIAamp ccfDNA/RNA Kit ($n = 4$). T-V, TRIzol-LS + QIAamp Viral RNA Kit ($n = 50$). CNA, QIAamp Circulating Nucleic Acid Kit ($n = 35$). ns, not significant; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. **e**, Association between haemolysis and cfRNA plasma concentration in healthy controls ($n = 40$). Amount of haemolysis was measured using optical density (OD) at 414 nm. Pearson and Spearman correlations are shown. **f**, Association between time in -80°C freezer and cfRNA plasma concentration in healthy controls ($n = 108$). **g**, Association between time in -80°C freezer and cfRNA plasma concentration across all cfRNA cohorts analysed in this study ($n = 237$). In box plots, the center line denotes the median, the box contains the interquartile range (IQR), and the whiskers denote extrema no farther than $1.5 \times \text{IQR}$.



Extended Data Fig. 2 | Optimization of RARE-Seq library preparation and capture.

a, Expression correlation from cfRNA collected in EDTA tubes or Streck RNA Complete tubes (average of $n = 3$ pairs). Log2NX, log2 normalized expression. **b**, Expression correlation from cfRNA extracted using the TRIzol LS + QIAamp Viral RNA method (T-V) and the QIAamp Circulating Nucleic Acid Kit (CNA) (average of $n = 3$ pairs). **c**, Expression correlation between cfRNA libraries generated using stranded and non-stranded methods (average of $n = 3$ pairs). **d**, Rarefaction analysis representing the relationship between total and unique sequencing depth for cfRNA libraries from **d**. The inset plot depicts the percent increase in unique sequencing depth for non-stranded libraries relative to paired stranded libraries. **e**, Expression correlation between cfRNA libraries generated with and without S1 nuclease end repair (average of $n = 3$ pairs). **f**, Rarefaction analysis representing the relationship between total and unique sequencing depth for cfRNA libraries from **e**. **g**, Estimated DNA

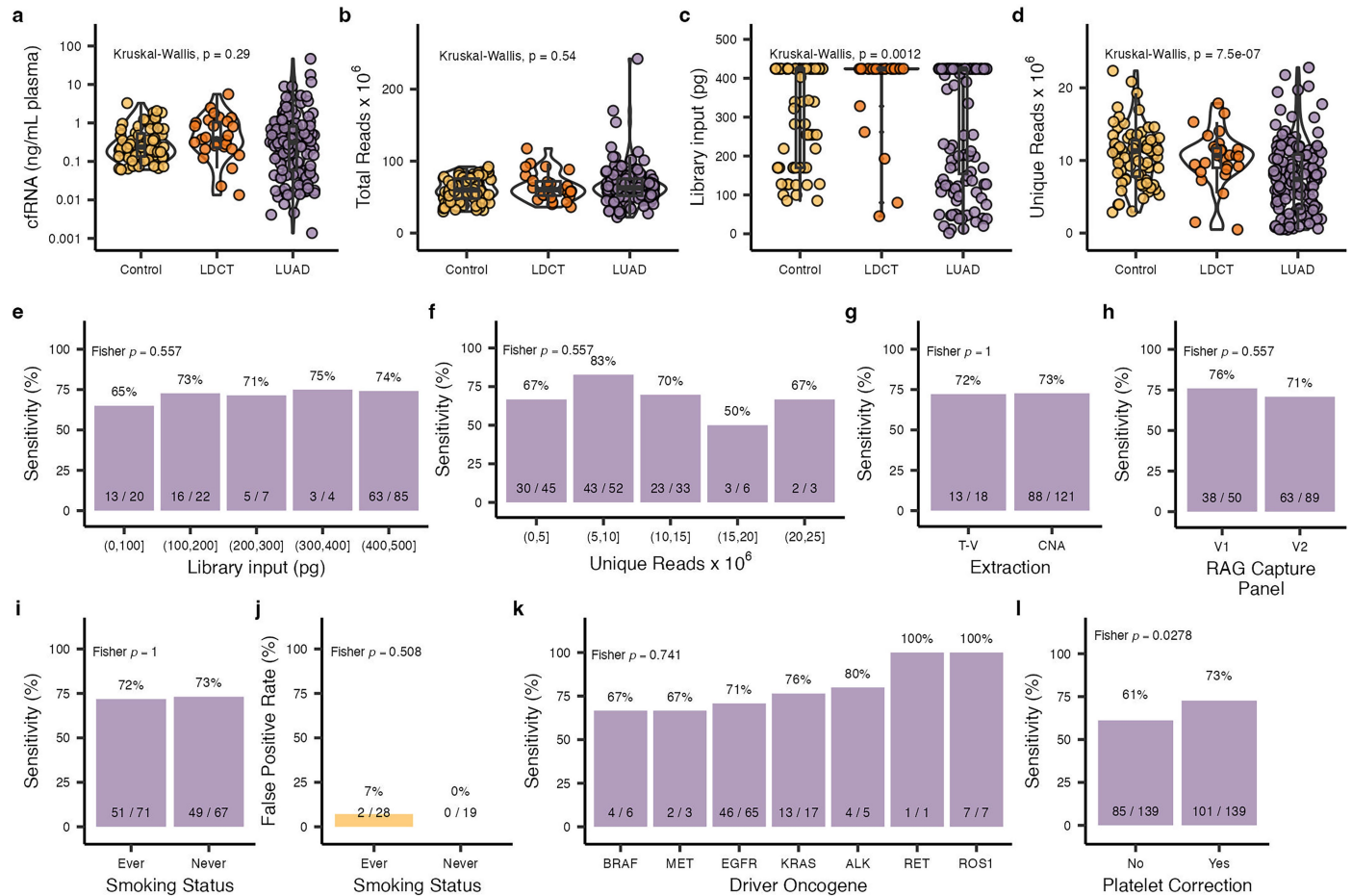
contamination after DNase I digestion performed either on or off Qiagen column ($n = 6$ per condition). Conditions were compared using a paired two-sided Student's *t*-test. **h**, Expression correlation between cfRNA samples digested on-column and off-column (average of $n = 6$ pairs). **i**, Distribution of RNA biotypes in cfRNA using SMART-Seq ($n = 3$). **j**, Expression correlation from cfRNA libraries generated using SMART-Seq and whole coding transcriptome RARE-Seq (median of $n = 3$ pairs). Histograms depict the log2NX distributions. The Venn diagram represents the number of coding genes with higher (or equivalent) log2NX in each method. Blue=coding genes, grey=non-coding genes. **k**, Heatmap showing pairwise Pearson correlation between all technical and biological replicates sequenced from a single individual ($n = 8$). In box plots, the center line denotes the median, the box contains the interquartile range (IQR), and the whiskers denote extrema no farther than $1.5 \times \text{IQR}$.



Extended Data Fig. 3 | Description and validation of capture panel targeting rare abundance genes.

a, Relationship between average gene expression in healthy cfRNA and overall percentage of healthy cfRNA samples with low or absent gene expression ($n = 50$). Red=rare abundance gene (RAGs; $n = 5,632$). Log2NX, log2 normalized expression. **b**, Relationship between average gene expression and expression stability measured by Gini coefficient in healthy cfRNA. Yellow=housekeeping genes ($n = 50$). **c**, Schematic depicting genes included in the core modules of the RAG-focused capture panel and examples of optional add-on modules that could be used to supplement RAG-focused capture panel. **d**, Cumulative frequency of non-small cell lung cancer (NSCLC) patients with one or more variants in genes added to the RAG-focused capture panel because they are recurrently altered or frequently aberrantly expressed in NSCLC ($n = 122$). Top 25 genes are shown. Variant data from the TCGA lung adenocarcinoma and lung squamous cell carcinoma project were downloaded

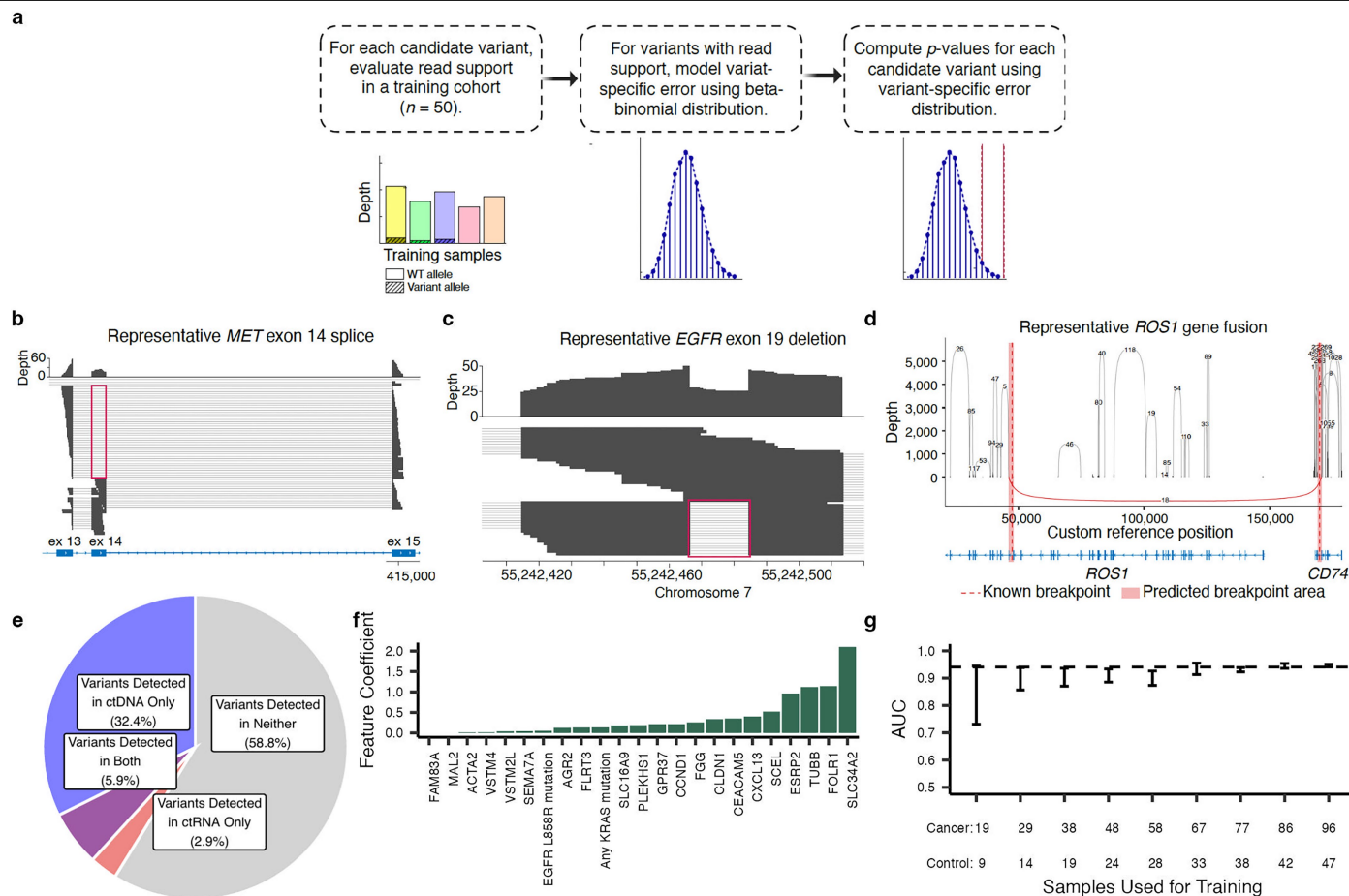
from cBioPortal. **e,f**, Expression correlation between whole coding transcriptome and RAG capture for **e**, all captured genes and **f**, housekeeping genes ($n = 5$ pairs). **g**, Schematic of platelet-directed correction approach using meta-reference controls with variable platelet expression. This approach was developed for samples captured with the RAG-focused panel which excluded platelet genes. See Methods for details. Z , z-score matrix. Z_p , z-score matrix of platelet genes. SVD, singular value decomposition. W , platelet correction factor. **h**, Association between platelet-specific gene expression and first principal component (PC1), after platelet-directed correction of cfRNA expression as depicted in **g**. Controls from Fig. 1d-k were used ($n = 32$). **i**, Association between NCI-H1975 spike detection and total sequencing depth. Logistic regression was used to estimate limit of detection (LOD95, shown as dashed lines) of the in silico spikes as described in Methods.



Extended Data Fig. 4 | Detection of lung adenocarcinoma ctRNA.

a–d, Relationship between sample library metrics from healthy controls ($n = 24$), low dose computed tomography controls (LDCT, $n = 26$), and lung adenocarcinoma (LUAD, $n = 50$) patients. Violin plots depict **a**, cfRNA concentration (ng per mL plasma), **b**, total sequencing depth, **c**, cfRNA mass used for library preparation, and **d**, unique sequencing depth. Each comparison was performed using Kruskal-Wallis test. **e–j**, Sensitivity of detection using LUAD cfRNA ES with respect to **e**, cfRNA mass used for library preparation, **f**, unique sequencing depth, **g**, extraction method used, **h**, capture panel used,

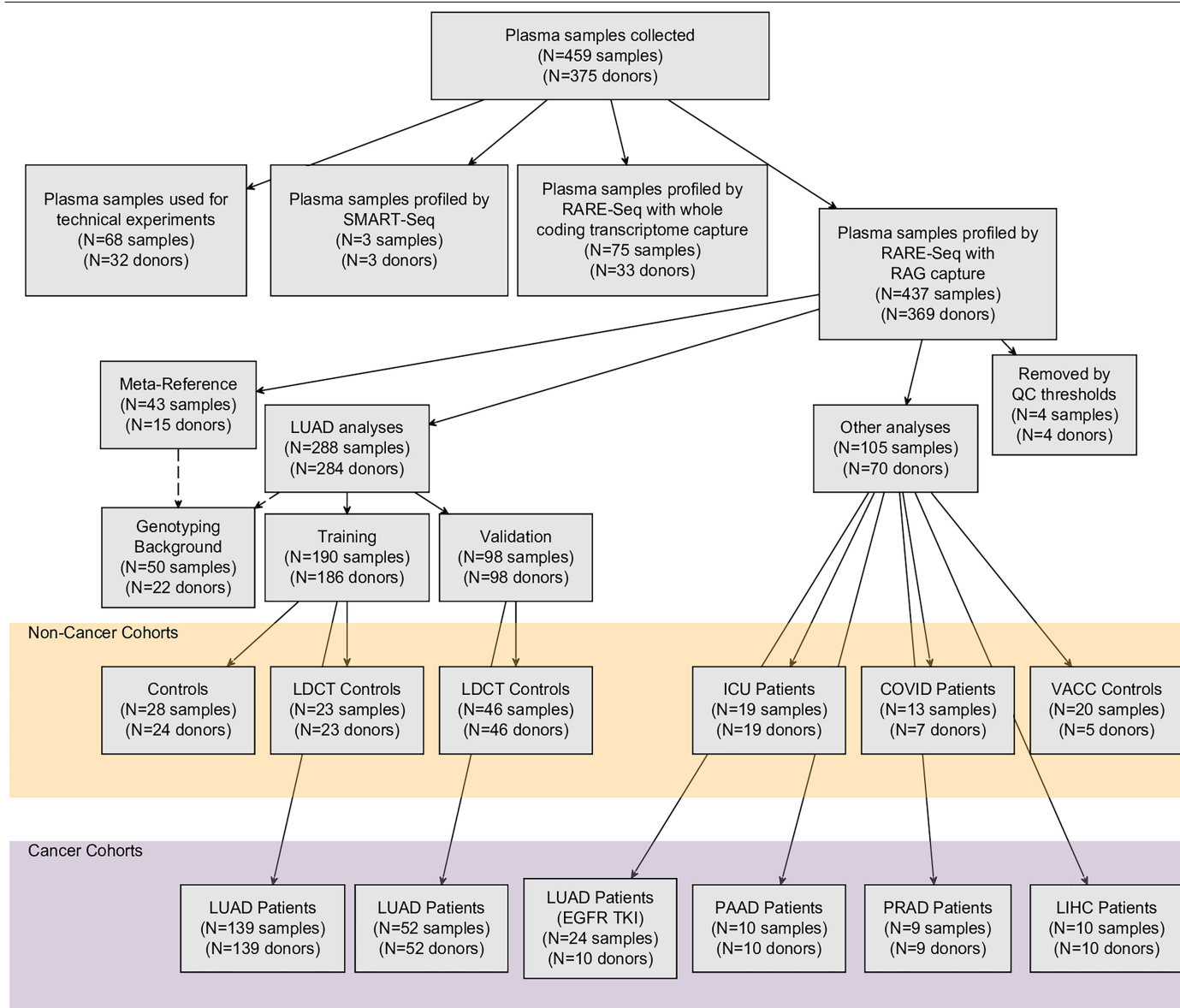
and **i**, patient smoking status. Detection threshold achieving $\geq 95\%$ specificity was used. **j**, False positive rate of LUAD cfRNA ES in controls with respect to smoking status ($n = 47$). **k**, Sensitivity of detection using LUAD cfRNA ES with respect to driver oncogene sub-type. **l**, Sensitivity of detection using LUAD cfRNA ES with respect to whether platelet correction was used. Comparison was performed using one-sided Fisher's exact test. In box plots, the center line denotes the median, the box contains the interquartile range (IQR), and the whiskers denote extrema no farther than $1.5 \times \text{IQR}$.



Extended Data Fig. 5 | Genotyping from lung adenocarcinoma ctRNA.

a, Schematic depicting ctRNA single nucleotide variant (SNV) calling approach (see Methods). **b,c**, Representative integrated genomics viewer (IGV) images displaying sequencing reads containing **b**, alternative splicing of the *MET* exon 14 and **c**, *EGFR* exon 19 deletion. **d**, Count of reads mapped to canonical splice junctions in *ROS1* and to fusion breakpoints for representative sample with tissue-adjudicated *ROS1/CD74* gene fusion. Junctions/breakpoints with ≥ 5 aligned reads are displayed. **e**, Pie chart summarizing the percentage of

tumour-adjudicated variants detected in ctDNA and/or ctRNA from plasma collected on the same day ($n = 34$ variants). **f**, LUAD elastic net (EN) model coefficients ($n = 23$ features total). **g**, Relationship between model training cohort size and 10-fold cross validated model performance. Data are presented as mean values $\pm$ SD of 10CV AUC at each sub-sampling level ($n = 10$ samples per level). The dotted line represents the 10-fold cross validated (10CV) AUC of the final LUAD EN model.



Extended Data Fig. 6 | Summary of cancer and non-cancer cfRNA cohorts.

LDCT, low-dose computed tomography. ICU, intensive care unit. COVID, COVID-19 infection. VACC, post-COVID-19 mRNA vaccination. LUAD, lung

adenocarcinoma. EGFR TKI, treated with epidermal growth factor receptor tyrosine kinase inhibitor. PAAD, pancreatic adenocarcinoma. PRAD, prostate adenocarcinoma. LIHC, liver hepatocellular carcinoma.

Extended Data Table 1 | Non-cancer cohort demographics

	Control		LDCT			ICU	COVID	VACC
	Meta-Reference	Training	Genotyping Background	Training	Validation	Other	Other	Other
No. of samples	43	28	7	23	46	19	13	20
No. of donors	15	24	7	23	46	19	7	5
Age (years)	29 (23-52)	30 (24-48)	60 (53-74)	62 (56-76)	63 (50-78)	65 (32-80)	56 (24-69)	45 (24-51)
Sex								
Male	8 (53.3%)	20 (83.3%)	5 (71.4%)	21 (91.3%)	25 (54.3%)	9 (47.4%)	4 (57.1%)	3 (60.0%)
Female	7 (46.7%)	4 (16.7%)	2 (28.6%)	2 (8.7%)	21 (45.7%)	10 (52.6%)	3 (42.9%)	2 (40.0%)
Smoking Status								
Ever	2 (13.3%)	2 (8.3%)	7 (100.0%)	23 (100.0%)	46 (100.0%)	10 (52.6%)	2 (28.6%)	
Never	13 (86.7%)	19 (79.2%)				9 (47.4%)	3 (42.9%)	5 (100.0%)
Unknown	0 (0.0%)	3 (12.5%)					2 (28.6%)	
Pack Years	0.0 (0.0)	0.0 (0.0)	28.8 (9.4)	40.4 (14.4)	33.7 (11.3)	0.0 (0.0)	6.7 (9.5)	0.0 (0.0)
Institution								
Stanford University	15	24	3	23	0	19	7	5
Massachusetts General Hospital			4	0	46			

LDCT, low-dose computed tomography. ICU, intensive care unit. COVID, COVID-19 infection. VACC, post-COVID-19 mRNA vaccination.

Article

Extended Data Table 2 | Cancer cohort demographics

	LUAD		LUAD (TKI)	PAAD	PRAD	LIHC
	Training	Validation	Other	Other	Other	Other
No. of samples	139	52	24	10	9	10
No. of donors	139	52	10	10	9	10
Age (years)	67 (38-87)	67 (30-92)	60 (31-77)	64 (42-82)	74 (54-86)	62 (41-82)
Sex						
Male	58 (41.7%)	19 (36.5%)	2 (20.0%)	4 (40.0%)	9 (100.0%)	6 (60.0%)
Female	81 (58.3%)	31 (59.6%)	8 (80.0%)	6 (60.0%)	0 (0.0%)	4 (40.0%)
Unknown	0 (0.0%)	2 (3.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Smoking Status						
Ever	71 (51.1%)	12 (23.1%)	3 (30.0%)	3 (30.0%)	3 (33.3%)	7 (70.0%)
Never	67 (48.2%)	38 (73.1%)	7 (70.0%)	7 (70.0%)	4 (44.4%)	3 (30.0%)
Unknown	1 (0.7%)	2 (3.8%)	0 (0.0%)	0 (0.0%)	2 (22.2%)	0 (0.0%)
Pack Years	13.1 (21.8)	1.3 (3.5)	1.0 (2.0)	1.0 (2.8)	7.3 (14.9)	26.0 (25.9)
Stage						
I	20 (14.4%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (30.0%)
II	8 (5.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	5 (50.0%)
III	15 (10.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (11.1%)	0 (0.0%)
IV	96 (69.1%)	52 (100.0%)	10 (100.0%)	10 (100.0%)	8 (88.9%)	2 (20.0%)
Institution						
Massachusetts General Hospital	0	52	10			
Memorial Sloan Kettering Cancer Center	40	0				
Stanford University	99	0		10	9	
University Hospital Zurich						10

LUAD, lung adenocarcinoma. TKI, treated with epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor. PAAD, pancreatic adenocarcinoma. PRAD, prostate adenocarcinoma. LIHC, liver hepatocellular carcinoma. A subset of LUAD samples were used in both the LUAD validation and the LUAD (TKI) cohorts (n=10).

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
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| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
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Software and code

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Data collection	Data processing was performed using fastp (v0.20.0), STAR (v2.7.0), RSEM (v.1.2.28), RNASEQC (v2.3.5), samtools (v1.11).
Data analysis	Data analysis was performed using R (v4.1.3). Expression normalization and platelet-directed correction used the R packages tximport (v1.22), edgeR (v3.36), RUVseq (v1.28). Differential expression analyses used the R packages DESeq2 (v1.34) and fgsea (v1.20). The elastic net framework employed the packages caret (v6.0), glmnet (v4.1), limma (v3.50.3), and pROC (v1.18). Variant calling analysis used samtools (v1.12) and STAR-Fusion (v1.6.0). The enrichment score framework and the variant calling framework also used custom R (v3.6.0, v4.1.3) and Python (v3.6.8) code that are available on github at https://github.com/RARESeq as well as http://rareseq.stanford.edu .

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De-identified clinical and demographic data for individuals considered in this study as well as gene signatures and associated enrichment scores, detected variants, and other relevant data are provided in the Supplementary Tables. Gene expression data and data relevant for somatic variant calling that were generated during this study are available at <http://rareseq.stanford.edu>. Gene expression data from GTEx, TCGA, and CCLE are publicly available from the UCSC Xena Repository (<https://xenabrowser.net/datapages/>). Gene expression data from additional SCLC tumors can be accessed using GEO (GSE60052). Cell type marker gene lists are publicly available from the PanglaoDB resource (<https://panglaoDB.se/markers.html>), and Molecular Signatures Database (MSigDb) gene lists are also publicly available (<https://www.gsea-msigdb.org/gsea/downloads.jsp>). Reference databases used in this study such as hg19 (https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.13) and GENCODE v27 (https://www.encodegenes.org/human/release_27.html) can be accessed using the provided weblinks.

Research involving human participants, their data, or biological material

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Reporting on sex and gender

The study design includes male and female participants. Sex was determined by patient electronic medical records at the respective enrollment center or by self-reporting. The meta-reference cohort was designed to have approximately equal representation of male (53%) and female (47%) participants. Summarized and disaggregated sex data are provided with the paper.

Reporting on race, ethnicity, or other socially relevant groupings

Socially constructed or socially relevant categorization was not collected for this study.

Population characteristics

Median age of participants included in this study was 67 (range 30-92) and 59 (range 23-80) for cancer patients and non-cancer controls, respectively. The sex of participants included in this study was 43% male and 65% male for cancer patients and non-cancer controls, respectively. Additional clinical and demographic details are provided with the paper.

Recruitment

Patients were recruited based on confirmed diagnosis of non-small cell lung cancer, pancreatic adenocarcinoma, prostate adenocarcinoma, hepatocellular carcinoma, acute respiratory distress syndrome, or COVID-19 disease, or based on negative low-dose computed tomography screening for lung cancer. All subjects were enrolled according to Institutional Review Board-approved protocols. The majority of samples were taken from our lab's biobank at Stanford University and so any selection biases are likely to reflect the patient population in our catchment area. Similar results from a validation cohort collected at an independent institution (Massachusetts General Hospital) suggest that any selection biases present do not substantially impact our findings.

Ethics oversight

All samples analyzed in this study were collected with informed consent on Institutional Review Board-approved protocols that complied with all relevant ethical regulations at their respective centers, including Stanford University, Massachusetts General Hospital, Memorial Sloan Kettering Cancer Center, and University Hospital Zurich.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Sample cohort sizes were determined by sample availability, so no statistical calculations were performed to determine cohort size.

Data exclusions

Three samples were excluded using quality control thresholds for sequencing depth and estimated DNA contamination rates as described in the "Mapping, deduplication, and quality control" section of Methods.

Replication

Technical experiments were performed with at least three biological replicates for each condition tested. In addition, study findings were successfully replicated in an external, independent sample cohort collected at a separate institution.

Randomization	Randomization was not applicable in this study because samples were assigned to groups based on confirmed diagnosis of non-small cell lung cancer, pancreatic adenocarcinoma, prostate adenocarcinoma, hepatocellular carcinoma, acute respiratory distress syndrome, or COVID-19 disease, or based on negative low-dose computed tomography screening for lung cancer.
Blinding	Blinding was not performed in this study. Blinding was not necessary for this retrospective, observational study for the purposes of method development and generating proof-of-concept data.

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☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

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Cell line source(s)	The NCI-H9175 cell line was acquired from American Type Culture Collection (ATCC CRL-5908). The MGH1157 and MGH170 cell lines were developed from tumor biopsies obtained from female patients enrolled on a Dana-Farber Harvard Cancer Center Institutional Review Board-approved protocol. The PC9/PC9-PERC17 cell lines were a generous gift from Steven Altschuler and Lani Wu at University of California San Francisco (PC9 originally acquired from the Minna Laboratory at UT Southwestern, per Ramirez et al Nat Commun 2016).
Authentication	NCI-H1975 was authenticated by confirming the presence of validated EGFR, TP53, PIK3CA, and CDKN2A variants in RNA sequencing data. MET-amplified cell lines were authenticated by MET fluorescence in situ hybridization (FISH) and by confirming sensitivity to combination of EGFR and MET tyrosine kinase inhibitors (TKIS).
Mycoplasma contamination	Tissue culture facilities were regularly tested for mycoplasma contamination and found to be negative.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used.

Plants

Seed stocks	<i>Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.</i>
Novel plant genotypes	<i>Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.</i>
Authentication	<i>Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.</i>